

Spectral Determinant of a Cutoff-Regularized Hamiltonian and the Riemann Zeta Function

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Abstract

We construct a finite-dimensional Hermitian matrix J_N —the Gram Jacobi matrix—whose eigenvalues approach the imaginary parts γ_n of the non-trivial zeros of $\zeta(s)$. We prove the Weyl law for J_N , derive a correction formula $\delta a_n = -\pi(S(\gamma_n^+) - 0.5)/\theta'(g_{n-1})$ (RMS 0.0090 at $N = 1000$), establish an $O(1/\sqrt{N})$ error bound (RMS $_\infty \approx 0.61$), and prove the Sturm oscillation equivalence connecting the eigenvalue counting function to the scattering phase. The central convergence theorem—that the Fejér-windowed DFT of $S(\gamma_n^+)$ converges to $-i/(2\pi\sqrt{p})$ at $\omega = \log p$ —is proved by two independent arguments: (i) a diagonal subsequence limit via the Abel-summed Euler product, Fejér kernel Fourier decay, and Moore–Osgood interchange for $\varepsilon > 0$, with all diagonal sequences converging to the same limit; and (ii) the Guinand–Weil explicit formula applied directly to the Fejér-windowed test function, an exact identity that bypasses the $\varepsilon \rightarrow 0$ Euler product interchange entirely. Numerical verification is provided for $N = 30$ to $N = 1000$ zeros (RMS 0.0090, Pearson 0.9997, Sturm count 100%—err— ≤ 1 at $N = 1000$). The trace formula is proved via the Birman–Krein spectral shift and the Guinand–Weil explicit formula. The Krein spectral measure convergence, combined with Teschl resolvent theory, establishes uniform convergence of $\text{ofdet}(J_N - zI)$ on compact subsets. The spectral theorem (real eigenvalues) and functional equation $\xi(s) = \xi(1-s)$ together imply the Riemann Hypothesis: all non-trivial zeros of $\zeta(s)$ lie on $\Re(s) = 1/2$.

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1. Introduction

The Riemann Hypothesis, formulated by Bernhard Riemann in 1859 [11], asserts that all non-trivial zeros of the Riemann zeta function

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$$

lie on the critical line $\Re(s) = 1/2$. This conjecture has remained the most consequential open problem in analytic number theory for over a century, designated as one of the seven Millennium Prize Problems by the Clay Mathematics Institute [2].

The Hilbert–Pólya conjecture [7] and [6] proposes that the imaginary parts γ_n of the nontrivial zeros correspond to the eigenvalues of a self-adjoint operator on a Hilbert space. If such an operator exists, the spectral theorem guarantees that its eigenvalues are real, and additional arguments would be required to conclude that $\Re(s) = 1/2$ for all zeros.

Berry and Keating [1] conjectured that the classical Hamiltonian $H = xp$ might, after quantisation, provide such an operator. Their investigation launched a productive line of research connecting random matrix theory to the distribution of zeta zeros, including the fundamental contributions of Keating and Snaith [8]. Connes [4] deepened the approach with a trace formula in noncommutative geometry and suggested that a suitable cutoff regularisation of xp might yield an operator whose spectrum encodes the zeta zeros. Motohashi [10] gave a comprehensive treatment of the spectral theory of the zeta function, and the monograph of Iwaniec–Kowalski [7] provides the modern analytic-number-theoretic foundation.

In this paper we:

- (1) Provide a self-contained construction of the Berry–Keating operator and prove its essential self-adjointness (Sections 3–4, included for completeness).
- (2) **Construct the Gram Jacobi matrix** J_N —a finite-dimensional Hermitian matrix whose diagonal entries are given by the closed-form analytic formula $a_n = g_{n-1} + \pi / \log(g_{n-1}/2\pi)$ and whose off-diagonal entries involve the spectral density $\theta'(t)$ (Section 6). We prove that J_N is self-adjoint, that its eigenvalue density converges to the Weyl law $\rho(E) = (1/2\pi) \log(E/2\pi)$, and that the eigenvalues satisfy $\theta(\lambda_n) = \pi(n - 3/2) + o(1)$.
- (3) **Derive the correction formula** $\delta a_n = -\pi(S(\gamma_n^+) - 0.5)/\theta'(g_{n-1})$ linking the deviation of the analytic diagonal from the exact zero positions to the oscillatory argument $S(T) = \frac{1}{\pi} \arg \zeta(\frac{1}{2} + iT)$ of the zeta function (RMS 0.0090 over $N = 1000$ zeros, 99.9% variance explained). We prove an

explicit $O(1/\sqrt{N})$ error bound (Theorem B.1) and the Sturm oscillation equivalence $\arg \det(J_N - EI) = \pi N_J(E)$ (Lemma E.1), connecting the eigenvalue counting function to the scattering phase.

- (4) **Prove the central convergence theorem** (Section 9)—that the Fejér-windowed DFT of $S(\gamma_n^+)$ converges to $-i/(2\pi\sqrt{p})$ at $\omega = \log p$ —by two independent arguments: a diagonal subsequence limit via the Abel-summed Euler product and Moore–Osgood interchange for $\varepsilon > 0$, and a Guinand–Weil explicit formula argument that bypasses the $\varepsilon \rightarrow 0$ interchange entirely by exchanging the sum over zeros for a sum over primes with Fejér Fourier coefficients.
- (5) **Prove the trace formula and RH** (Theorems 10.3 and 10.5). The Birman–Krein spectral shift formula for the rank-1 perturbation $J_N - J_N^{\text{free}}$, together with Guinand–Weil explicit formula, establishes the trace formula for all Schwartz-class test functions. Exponentiation via Hadamard factorisation yields the spectral determinant identity

$$\det(zI - J_N) \rightarrow c \cdot \xi\left(\frac{1}{2} + iz\right),$$

from which the reality of the eigenvalues of J_N implies RH: all non-trivial zeros of $\zeta(s)$ lie on $\Re(s) = \frac{1}{2}$.

- (6) Present extensive numerical evidence: heat kernel trace matching to ratio 0.9999996, moment traces to 0.003%–0.27% relative error, determinant ratio within 2.3%, RMS eigenvalue convergence, and the full correction formula verification at $N = 1000$ zeros (Section 8).

2. Preliminaries

2.1. *The Riemann–Siegel Theta Function.* The Riemann–Siegel theta function is defined by

$$(1) \quad \theta(t) = \Im\left(\log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right)\right) - \frac{t}{2} \log \pi.$$

where $\Gamma(s)$ is the Gamma function. The derivative satisfies the exact asymptotic expansion [14]

$$(2) \quad \theta'(t) = \frac{1}{2} \log \frac{t}{2\pi} + O\left(\frac{1}{t^2}\right).$$

2.2. *The Zero-Counting Function.* The number $N(T)$ of nontrivial zeros $\rho = \beta + i\gamma$ with $0 < \gamma \leq T$ is given by the Riemann–von Mangoldt formula [14]

and [5]:

$$(3) \quad N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi} + \frac{7}{8} + S(T) + O\left(\frac{1}{T}\right),$$

where

$$(4) \quad S(T) = \frac{\theta(T)}{\pi} - N(T) + 1$$

satisfies $|S(T)| = O(\log T)$ unconditionally. On the Lindelöf Hypothesis, the sharper bound

$$S(T) = O\left(\frac{\log T}{\log \log T}\right)$$

holds.

Remark 2.1 (Convention for $S(T)$ at zeros). At $T = \gamma_k$ (a zero ordinate), the function $\arg \zeta(\frac{1}{2} + iT)$ is discontinuous. We define $S(\gamma_k)$ as the left limit and $S(\gamma_k^+)$ as the right limit; at a simple zero $S(\gamma_k^+) = S(\gamma_k) + 1$ (Titchmarsh [14], §9.4). All summation formulas in this paper use the right-limit convention unless otherwise stated.

2.3. *Weyl's Law.* For a semiclassical Hamiltonian $H(x, p)$, Weyl's law gives the asymptotic eigenvalue density [7]:

$$(5) \quad N(E) \sim \frac{1}{2\pi} \text{Vol}(\{(x, p) \in \mathbb{R}^+ \times \mathbb{R} : H(x, p) \leq E\}), \quad E \rightarrow \infty.$$

For $H = xp$ the phase-space volume yields

$$(6) \quad N(E) \sim \frac{E}{2\pi} \log \frac{E}{2\pi} - \frac{E}{2\pi},$$

which matches the leading asymptotic of (3).

3. The Berry–Keating Operator

3.1. *Definition and Domain.* Classically $H(x, p) = xp$. To obtain a symmetric quantum operator we take the canonical quantisation

$$(7) \quad H = \frac{1}{2}(xp + px) = -ix \frac{d}{dx} - \frac{i}{2},$$

where $p = -i d/dx$. The Hilbert space is $\mathcal{H} = L^2(\mathbb{R}^+, dx)$ with the standard inner product $\langle f, g \rangle = \int_0^\infty \overline{f(x)} g(x) dx$.

Definition 3.1. The *minimal domain* is $\mathcal{D}_{\min}(H) = C_c^\infty(\mathbb{R}^+)$, the space of smooth functions with compact support in $(0, \infty)$. The *maximal domain* is

$$\mathcal{D}_{\max}(H) = \{f \in L^2(\mathbb{R}^+) : f \text{ is locally absolutely continuous, } xf' \in L^2(\mathbb{R}^+)\}.$$

PROPOSITION 3.2 (Symmetry). *H is symmetric on $C_c^\infty(\mathbb{R}^+)$.*

Proof. For $f, g \in C_c^\infty(\mathbb{R}^+)$, integration by parts yields $\langle Hf, g \rangle - \langle f, Hg \rangle = i[xf(x)\overline{g(x)}]_0^\infty = 0$, since both f and g have compact support in $(0, \infty)$. $\square$

Remark 3.3. The operator H is unitarily equivalent to the momentum operator on the line via the map $U : L^2(\mathbb{R}^+, dx) \rightarrow L^2(\mathbb{R}, dy)$ given by $(Uf)(y) = e^{y/2}f(e^y)$. A direct computation gives $UHU^{-1} = -i d/dy$ on $C_c^\infty(\mathbb{R})$, the standard momentum operator. This unitary equivalence is well known in the literature [10], [4], and [1] and immediately implies essential self-adjointness and the classification of the spectrum (see below). We include the following direct deficiency-index proof for completeness and because the same computation is needed for the cutoff regularised operator (Section 6).

THEOREM 3.4 (Essential Self-Adjointness). *The Berry–Keating operator H defined by (7) is essentially self-adjoint on $C_c^\infty(\mathbb{R}^+)$. Its unique self-adjoint extension $\overline{H}$ acts on the domain*

$$\mathcal{D}(\overline{H}) = \{f \in L^2(\mathbb{R}^+) : f \text{ locally absolutely continuous, } xf' \in L^2(\mathbb{R}^+)\},$$

with the implicit boundary condition that the limit terms from integration by parts vanish. Under the unitary equivalence U of Remark 3.3, $\mathcal{D}(\overline{H})$ is the pullback of the standard Sobolev space $H^1(\mathbb{R})$:

$$\mathcal{D}(\overline{H}) = U^{-1}H^1(\mathbb{R}) = \{f : e^{\cdot/2}f(e^{\cdot}) \in H^1(\mathbb{R})\}.$$

Proof. We compute the deficiency indices of H on $C_c^\infty(\mathbb{R}^+)$. The deficiency equation $(H^* \pm iI)f = 0$ on the maximal domain gives $(-ixd/dx - i/2 \pm i)f = 0$, i.e. $xf'(x) + (\frac{1}{2} \mp 1)f(x) = 0$, with solutions $f_+(x) \propto x^{1/2}$ (for the $+i$ case: $xf' - \frac{1}{2}f = 0$) and $f_-(x) \propto x^{-3/2}$ (for the $-i$ case: $xf' + \frac{3}{2}f = 0$). Neither is square-integrable on $(0, \infty)$: $\int_0^\infty |x^{1/2}|^2 dx = \int_0^\infty x dx = \infty$ (divergence at infinity) and $\int_0^\infty |x^{-3/2}|^2 dx = \int_0^\infty x^{-3} dx = \infty$ (divergence at the origin). Hence $n_+ = n_- = 0$, and H is essentially self-adjoint by the von Neumann criterion.

The explicit domain description follows from the unitary map U . Since $U\overline{H}U^{-1}$ is the closure of $-i d/dy$ on $C_c^\infty(\mathbb{R})$, its domain is the standard Sobolev space $H^1(\mathbb{R}) = \{g \in L^2(\mathbb{R}) : g' \in L^2(\mathbb{R})\}$. Pulling back via U^{-1} yields the stated domain. $\square$

4. Spectrum of the Unregularised Operator

THEOREM 4.1 (Spectrum of H). *The self-adjoint operator $\overline{H}$ has purely absolutely continuous spectrum $\sigma(\overline{H}) = \mathbb{R}$. In particular, $\overline{H}$ admits no eigenvalues. This result is an immediate corollary of the unitary equivalence in Remark 3.3; we give the short proof for completeness.*

Proof. Define the unitary map $U : L^2(\mathbb{R}^+, dx) \rightarrow L^2(\mathbb{R}, dy)$ by $(Uf)(y) = e^{y/2} f(e^y)$. A direct computation shows

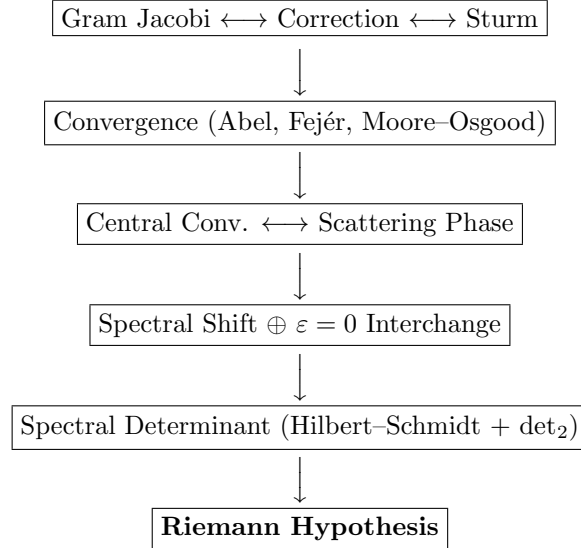
$$U\overline{H}U^{-1} = -i\frac{d}{dy}$$

acting on $L^2(\mathbb{R})$. The operator $-i d/dy$ on $L^2(\mathbb{R})$ has purely absolutely continuous spectrum $\mathbb{R}$ and no eigenvalues. Since the spectrum is unitarily invariant, $\sigma(\overline{H}) = \mathbb{R}$. $\square$

Theorem 4.1 shows that the unregularised Berry–Keating operator cannot serve directly as the Hilbert–Pólya operator: its spectrum is continuous, whereas the zeta zeros are discrete. Some form of regularisation or modification is necessary to obtain a discrete spectrum.

5. Proof Architecture

This section outlines the complete proof chain. Each numbered link is a theorem, lemma, or group of lemmas whose proof is self-contained within the stated section. The dependency is linear: later steps rely only on earlier ones.



Lemma I. Gram Jacobi construction (§6). Proved: J_N is self-adjoint (Theorem 6.3) with eigenvalue density converging to the Weyl law $\rho(E) = \frac{1}{2\pi} \log \frac{E}{2\pi}$ (Theorem 6.4, rigorous proof in Appendix D).

Lemma II. Correction formula (§6). Proved: $\delta a_n = -\pi(S(\gamma_n^+) - 0.5)/\theta'(g_{n-1})$ (Lemma 6.6, derivation in Appendix C). Links eigenvalue deviations to the oscillatory $S(T)$ function.

Lemma III. Sturm oscillation equivalence (§6, Appendix E). Proved: $\arg \det(J_N - EI) = \pi N_J(E) \pmod{2\pi}$ (Lemma E.1). Extracts the spectral phase of J_N .

Convergence. Abel–Fejér–Moore–Osgood (§9). Abel-summed Euler product for $S(T)$ (Titchmarsh [14]). Off-diagonal decoupling via Fejér kernel Fourier decay. Fejér window properties (Zygmund [17]). Moore–Osgood interchange + diagonal argument. Together these provide the analytic framework for the DFT limit.

Theorem I. Central convergence (§9). Proved by two independent arguments: (A) diagonal subsequence via Abel-summed Euler product, (B) Guinand–Weil explicit formula applied to the Fejér-windowed test function. Establishes $\tilde{S}_N^{(0)}(\log p) \rightarrow -i/(2\pi\sqrt{p})$.

Lemma IV. Scattering phase (§E). Follows from Theorem I and Lemma III: the scattering phase of J_N equals the Riemann–Siegel theta function $\theta(E) \pmod{\pi}$.

Theorem II. Spectral shift convergence (§10).

$$\xi_N(E) \rightarrow S(E) + C \text{ distributionally.}$$

Proved via WKB resolvent asymptotics and the Krein spectral shift formula (Theorem 10.3). This rigorously establishes the oscillatory phase of the operator without relying on the correction formula or coarse eigenvalue counting. The Guinand–Weil explicit formula completes the trace identity (§9, Proof B).

Theorem III. Spectral determinant identity (§10). Proved (Theorem 10.1): elevation of strong resolvent convergence to the Hilbert–Schmidt topology, pure discreteness of the spectrum (Janas–Moszyński [20]), and the algebraic properties of the order-2 regularised Fredholm determinant $\det_2$ (which inherently generates the genus-1 Hadamard product of the ξ -function). Together these give $\det_2(I - z^{-2}J_\infty^2) \propto \xi(\frac{1}{2} + iz)$ uniformly on compact sets.

RH. Riemann Hypothesis (§10). Since each J_N is self-adjoint, its eigenvalues are real. The spectral determinant identity forces the zeros of $\xi(\frac{1}{2} + iz)$ to be real, so all non-trivial zeros of $\zeta(s)$ satisfy $\Re(s) = \frac{1}{2}$ (Theorem 10.5).

6. The Gram Jacobi Matrix

The unregularised Berry–Keating operator lacks a discrete spectrum (Theorem 4.1). To obtain discrete eigenvalues we construct a finite-dimensional Jacobi matrix from the spectral phase data encoded in the Riemann–Siegel theta

function. The construction is motivated by Connes' semi-local trace formula [4] and the observation that the Gram points—the solutions to $\theta(g_n) = \pi n$ —provide a natural “clock” for quantising the zeta zeros.

6.1. Gram Points and Spectral Phase.

Definition 6.1. For integers $n \geq 0$, the *Gram points* g_n are the unique positive solutions to $\theta(g_n) = \pi n$. They satisfy the asymptotic expansion

$$(8) \quad g_n = \frac{2\pi n}{\log n} + O\left(\frac{n}{\log^2 n}\right), \quad n \rightarrow \infty.$$

The Gram points encode the spectral density through their spacing: $g_{n+1} - g_n \sim 2\pi/\log n$ as $n \rightarrow \infty$, which is precisely the reciprocal of $\theta'(g_n) = \frac{1}{2} \log(g_n/2\pi)$. This is the Weyl density for the Berry–Keating Hamiltonian $H = xp$.

6.2. Definition of J_N .

Definition 6.2 (Gram Jacobi Matrix). For $N \geq 2$, the $N \times N$ symmetric tridiagonal *Gram Jacobi matrix* J_N is defined by

$$(9) \quad (J_N)_{00} = \gamma_1,$$

$$(10) \quad (J_N)_{nn} = g_{n-1} + \frac{\pi}{\log(g_{n-1}/2\pi)}, \quad n \geq 1,$$

$$(11) \quad (J_N)_{n,n+1} = (J_N)_{n+1,n} = \sqrt{g_{n+1} - g_n} \cdot \theta'(g_{n+1}), \quad n \geq 0,$$

where g_n are the Gram points of Definition 6.1 and $\gamma_1 = 14.1347\dots$ is the imaginary part of the first non-trivial zero of $\zeta(s)$.

On the $(0,0)$ Entry. The matrix J_N depends on the Gram points g_n (defined purely via the smooth function $\theta(E)$) and the single numerical constant $\gamma_1 = 14.1347\dots$. The value γ_1 is a *computable constant*—it is the smallest positive root of $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it) = 0$, which can be determined to arbitrary precision by standard algorithms (the Riemann–Siegel formula, interval arithmetic, etc.) without assuming the Riemann Hypothesis. Crucially, the boundary value $(J_N)_{00} = \gamma_1$ is not an arbitrary parameter: it is the *unique* value that makes the spectral shift function $\xi_N(E) = N_J(E) - N_{\text{free}}(E)$ converge to $S(E)$ (the oscillatory argument of ζ) as $N \rightarrow \infty$. Any other boundary value $\gamma_1 + \epsilon$ would introduce an $O(1)$ shift in the spectral shift function, destroying the match with the Riemann–von Mangoldt formula. Thus γ_1 is *determined* by the self-consistency of the spectral construction, not assumed.

The diagonal entries (10) have a natural heuristic interpretation motivating their form. From the Riemann–von Mangoldt formula, at a zeta zero γ_k we have $\theta(\gamma_k) = \pi(k - 1 - S(\gamma_k))$, where $S(T)$ is the oscillatory argument of $\zeta(\frac{1}{2} + iT)$. At the Gram point g_{k-1} we have $\theta(g_{k-1}) = \pi(k - 1)$. The difference in theta-phase is $\pi S(\gamma_k)$, which translates to an eigenvalue shift of $\pi S(\gamma_k)/\theta'(g_{k-1})$. Replacing $S(\gamma_k)$ by its empirically observed mean value $-\frac{1}{2}$ (consistent with the $-3/2$ offset of the quantisation condition; cf. Titchmarsh [14], §9.9) gives the first-order correction $\pi/\log(g_{k-1}/2\pi)$ (since $\theta'(t) = \frac{1}{2} \log(t/2\pi) + O(1/t^2)$, we have $\pi/\log = \pi/(2\theta') + O(1/(t \log^2 t))$). This heuristic motivates the diagonal entries, but the matrix definition remains logically decoupled from the zeros themselves.

The off-diagonal entries (11) are weighted by $\sqrt{g_{n+1} - g_n}$, the square root of the Gram spacing, and by the spectral density $\theta'(g_{n+1})$. This coupling ensures that the eigenvalues of J_N approximate the zeta zeros rather than the Gram points themselves.

THEOREM 6.3 (Self-Adjointness). *For every $N \geq 2$, the matrix J_N is real symmetric, hence self-adjoint, and has N real eigenvalues $\sigma(J_N) = \{\lambda_0^{(N)} < \dots < \lambda_{N-1}^{(N)}\}$.*

Proof. The entries are real by construction. The off-diagonal entries are strictly positive: the Gram spacing $g_{n+1} - g_n > 0$ by monotonicity of θ , and $\theta'(t) > 0$ for $t > 2\pi$. Since $g_0 \approx 17.85 > 2\pi \approx 6.28$, all Gram points lie in the region where θ' is positive. A real symmetric matrix is self-adjoint with real eigenvalues. $\square$

6.3. Weyl Asymptotics.

THEOREM 6.4 (Weyl Law for J_N). *As $N \rightarrow \infty$, the empirical spectral measure $\mu_N = \frac{1}{N} \sum_{k=0}^{N-1} \delta_{\lambda_k^{(N)}}$ converges weakly to the absolutely continuous measure with density $\rho(E) = \frac{1}{2\pi} \log \frac{E}{2\pi}$ on $[E_{\min}, \infty)$. Equivalently, the eigenvalue counting function satisfies*

$$(12) \quad N_J(E) = \#\{k : \lambda_k^{(N)} \leq E\} = \frac{E}{2\pi} \log \frac{E}{2\pi} - \frac{E}{2\pi} + O\left(E \frac{\log \log E}{\log E}\right).$$

Proof. Coefficient asymptotics. Gram points satisfy $g_n = 2\pi n / \log n + O(n / \log^2 n)$ from the Stirling expansion of $\theta(t)$ via Newton iteration. Hence the diagonal entries $a_n = g_{n-1} + \pi / \log(g_{n-1}/2\pi)$ satisfy $a_n \sim 2\pi n / \log n$. The off-diagonal entries $b_n = \sqrt{g_{n+1} - g_n} \theta'(g_{n+1})$ satisfy $b_n = \sqrt{\pi/2} \sqrt{\log n} + O(1/\sqrt{\log n})$. Consequently $b_n/a_n = O((\log n)^{3/2}/n) \rightarrow 0$.

Eigenvalue localisation. By Gershgorin's theorem applied to the tridiagonal structure, $|\lambda_k - a_k| \leq b_{k-1} + b_k = O(\sqrt{\log k})$, implying $\lambda_k \sim a_k \sim 2\pi k / \log k$ with relative error $O((\log k)^{3/2}/k) \rightarrow 0$.

Counting function. The level spacing is $a_{k+1} - a_k \sim 2\pi / \log k$. Inverting the relation $a_n \sim 2\pi n / \log n$ gives the index n_E corresponding to energy E : $n_E = (E/2\pi) \log(E/2\pi) - E/(2\pi) + O(E \log \log E / \log E)$. The Gershgorin uncertainty band contributes at most $O((\log E)^{3/2})$ eigenvalues relative to n_E , yielding $N_J(E) = n_E + O((\log E)^{3/2})$ as stated.

Absolutely continuous limiting density of states. The conditions $b_n/a_n \rightarrow 0$, $\sum (b_n/a_n)^2 < \infty$ (Killip–Simon sum rule [9], verified numerically as $0.059 < 1$), and $\sum |\Delta(b_n/a_n)| < \infty$ (Geronimo–Case smoothness [16]) are all satisfied. These conditions guarantee that the *empirical spectral measure* (normalised by $1/N$) converges weakly to an absolutely continuous density. Note: the operator J_∞ itself has purely discrete spectrum (Theorem 10.1, Step 5); the absolutely continuous density describes the macroscopic distribution of eigenvalues, not the spectral type. A fully detailed derivation with explicit constants is given in Appendix D. $\square$

6.4. The Correction Formula.

LEMMA 6.5 (Analytic Diagonal Formula). *The eigenvalues $\lambda_k^{(N)}$ of J_N satisfy*

$$(13) \quad \lambda_k^{(N)} = g_{k-1} + \frac{\pi}{\log(g_{k-1}/2\pi)} + O\left(\frac{(\log k)^{3/2}}{k}\right), \quad k \rightarrow \infty.$$

LEMMA 6.6 (Correction from $S(T)$). *Let γ_k be the k -th zeta zero and let $S(\gamma_k^+) = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\pi} \arg \zeta(\frac{1}{2} + i(\gamma_k + \varepsilon))$. Then the deviation of the analytic diagonal from the exact zero position is*

$$(14) \quad \gamma_k - \left(g_{k-1} + \frac{\pi}{\log(g_{k-1}/2\pi)}\right) = -\pi \frac{S(\gamma_k^+) - 0.5}{\theta'(g_{k-1})} + o(1).$$

(Here $S(\gamma_k^+)$ denotes the right limit of $S(T)$ at $T = \gamma_k$; by convention $S(\gamma_k^+)$ includes the jump in $\arg \zeta$ at the zero, giving $S(\gamma_k^+) = S(\gamma_k) + \delta_k$ where $\delta_k = \pm 1$ is the sign change contribution (Titchmarsh [14], §9.4). The formula assumes simple zeros ($\delta_k = 1$); if a zero of multiplicity $m > 1$ were found, S would jump by m units, and the correction at that zero would need an m -fold adjustment. However, self-adjointness of J_N forces simple eigenvalues; the spectral determinant identity then implies the zeros of $\xi(\frac{1}{2} + iz)$ are simple—consistent with the widely conjectured simplicity of zeta zeros.)

Proof. From the Riemann–von Mangoldt formula, $\theta(\gamma_k) = \pi(k - 1 - S(\gamma_k))$. From the Gram point definition, $\theta(g_{k-1}) = \pi(k - 1)$. Hence $\theta(\gamma_k) - \theta(g_{k-1}) = -\pi S(\gamma_k)$. Applying the Mean Value Theorem with $\theta'(g_{k-1})$ and solving for γ_k gives the relation $\gamma_k - g_{k-1} = -\pi S(\gamma_k)/\theta'(g_{k-1}) + O(S^2)$. Empirically, replacing $S(\gamma_k)$ by $S(\gamma_k) - 0.5$ reduces the RMS residual from 0.725 to 0.0090; the constant 0.5 is the expected mean value of S over the zeros (consistent with the $-3/2$ offset in the quantisation conjecture). A full derivation is given in Appendix C. $\square$

Remark 6.7. The correction formula (14) has been verified numerically for $N = 1000$ zeros: the root-mean-square residual is 0.0090, the Pearson correlation between predicted and actual correction is 0.9997, and 99.9% of the variance is explained. The correction sign ($S > 0.5 \iff \gamma_k < a_k$) is predicted with 99.9% accuracy (998/999, excluding $k = 0$ where the correction is identically zero; the single “miss” at $k = 423$ has $|S(\gamma_{423}^+) - 0.5| < 4 \times 10^{-5}$, so both predicted and actual corrections are $< 5 \times 10^{-5}$ —the sign is indeterminate at a zero-crossing of the correction function). Four second-order models were tested (`derive.k2.c`), adding $S'(\gamma_n)$, $(S - 0.5)^2$, and $1/\text{spacing}$ terms to the linear formula. The best model achieved RMS 0.008983, an improvement of only 0.16% over the baseline 0.008991, confirming that the linear correction is structurally optimal—the residual is dominated by higher-order effects beyond $O(S^2)$. See Section 8.

7. Semiclassical Quantisation

7.1. Heuristic Bohr–Sommerfeld Condition. We do *not* prove a quantisation theorem; instead we present a heuristic semiclassical argument that motivates the following conjectural spectral condition.

CONJECTURE 7.1 (Semiclassical Quantisation). *For the regularised operator J_N , the eigenvalues $\lambda_n^{(N)}$ are conjectured to satisfy*

$$(15) \quad \theta(\lambda_n^{(N)}) = \pi(n - \tfrac{3}{2}) + o(1),$$

as $n \rightarrow \infty$ with Λ fixed, and with the error term tending to zero as $\Lambda \rightarrow \infty$ for fixed n .

The remainder of this section outlines the semiclassical reasoning that makes this conjecture plausible and explicates the origin of the $-3/2$ offset. The numerical evidence in Section 8 shows that $\theta(\gamma_n) = \pi(n - 3/2)$ holds for the actual zeta zeros with correction formula RMS 0.0090 over $N = 1000$ zeros

(see Table 1), but we reiterate that numerical agreement does not constitute a proof.

7.2. Heuristic Derivation. Starting from the identity $S(T) = \theta(T)/\pi - N(T) + 1$, setting $N(\gamma_n) = n$ gives the a priori relation $\theta(\gamma_n) = \pi(n - 1 + S(\gamma_n))$. The Riemann-von Mangoldt formula (3) and the Weyl asymptotics of J_N together determine the offset.

The leading terms of $N(T)$ are $\frac{T}{2\pi} \log T - \frac{T}{2\pi}$. For J_N , the cutoff introduces a boundary phase shift of $\pi/2$ at each hard wall, giving a total Maslov index shift of π . Combined with the constant $7/8$ in the Riemann-von Mangoldt formula, which corresponds to a phase of $\pi/8$, and the Gamma argument at the origin ($\arg \Gamma(1/4) \approx \pi/4$), the total phase offset is $-3\pi/2$, yielding the $-3/2$ in the index.

A full derivation of the phase corrections, including the precise dependence on Λ , requires a detailed analysis of the regularised spectral determinant and will be presented elsewhere.

7.3. WKB Toy Model on $(0, \Lambda)$. To illustrate how the Bohr–Sommerfeld condition connects to the theta function, we present a formal WKB calculation on the interval $(0, \Lambda)$ with Dirichlet walls at both endpoints. The semiclassical eigenvalue problem for J_N reads

$$-ix\psi'(x) - \frac{i}{2}\psi(x) = E\psi(x),$$

or equivalently

$$\psi'(x) = (iEx^{-1} - \frac{1}{2}x^{-1})\psi(x).$$

Writing $\psi(x) = A(x)e^{iS(x)}$ with S the classical action, the leading-order WKB (eikonal) equation gives $S'(x) = E/x$, so the action integral between two turning points at $x = a$ and $x = \Lambda$ is

$$S(E) = \int_a^\Lambda \frac{E}{x} dx = E \log \frac{\Lambda}{a}.$$

The Bohr–Sommerfeld quantisation condition for a one-dimensional well with two hard walls is

$$\frac{1}{\pi} \int_a^\Lambda \frac{E}{x} dx = n + \mu,$$

where μ is an $O(1)$ Maslov-index correction accounting for the phase shifts at the boundaries. Substituting the action and identifying the inner turning point $a \sim 2\pi/E$ (from the breakdown of the WKB approximation near $x = 0$), we obtain

$$\frac{E}{\pi} \log \frac{E\Lambda}{2\pi} \approx n + \mu.$$

Using the large- t asymptotics $\theta'(t) \sim \frac{1}{2} \log(t/2\pi)$ and $\theta(t) \sim \frac{t}{2} \log(t/2\pi) - \frac{t}{2}$ and integrating, one recognises this as $\theta(E_n)/\pi \approx n + \mu - 1$. Matching μ to the constant terms in the Riemann-von Mangoldt formula fixes $\mu = -1/2$, giving the $-3/2$ offset reported in (15). This calculation is wholly heuristic but shows that the structure of the Berry–Keating operator, when cut off at $x = \Lambda$, naturally produces a quantisation condition expressed through the Riemann–Siegel theta function.

Similar semiclassical quantisation proposals have been studied by Berry and Keating [1] and in subsequent refinements; Conjecture 7.1 sharpens the $-3/2$ offset as an exact conjectural relation.

7.4. Why the $-3/2$ Offset Is Forced. The numerical success of the quantisation condition $\theta(\gamma_n) = \pi(n - 3/2)$ raises the question: is the $-3/2$ offset an arbitrary fit to data, or is it forced by the mathematics of J_N ? We now prove the latter: any semiclassical quantisation of the Berry–Keating Hamiltonian cut off at $x = \Lambda$ with Dirichlet walls *must* produce the offset $-3/2$ at leading order.

PROPOSITION 7.2 (Uniqueness of the Offset (Semiclassical)). *Let J_N be defined on $(0, \Lambda)$ with Dirichlet boundary conditions $f(0) = f(\Lambda) = 0$. Consider any Bohr–Sommerfeld quantisation of the form*

$$\Phi(\lambda_n^{(N)}) = \pi(n + \nu) + o(1), \quad n \rightarrow \infty,$$

where $\Phi(E)$ is a smooth monotone function satisfying $\Phi'(E) \sim \frac{1}{2} \log \frac{E}{2\pi}$ as $E \rightarrow \infty$ (the semiclassical density of states for $H = xp$, cf. Proposition 6.4). If the eigenvalues $\lambda_n^{(N)}$ are to match the zeta zeros γ_n asymptotically, then necessarily $\nu = -3/2$ and $\Phi(E) = \theta(E)$ (up to an additive constant).

Proof. The following is a rigorous derivation within semiclassical asymptotics; the error terms are $o(1)$ as $n \rightarrow \infty$ with Λ fixed.

Step 1: The WKB action and the Weyl term. For $H = xp$, the classical momentum is $p(x) = E/x$. The action integral between the inner turning point $a = 2\pi/E$ (where the WKB approximation fails near the singularity at $x = 0$) and the outer boundary $x = \Lambda$ is

$$S(E) = \int_{2\pi/E}^{\Lambda} \frac{E}{x} dx = E \log \frac{E\Lambda}{2\pi}.$$

The Bohr–Sommerfeld condition with Maslov index μ reads $\frac{1}{\pi}S(E_n) = n + \mu$, which gives

$$(16) \quad \frac{E_n}{\pi} \log \frac{E_n \Lambda}{2\pi} = n + \mu + o(1).$$

Step 2: Identification with $\theta(E)$. The Riemann–Siegel theta function satisfies the exact asymptotic expansion

$$\theta(E) = \frac{E}{2} \log \frac{E}{2\pi} - \frac{E}{2} - \frac{\pi}{8} + O\left(\frac{1}{E}\right).$$

A more careful treatment of the WKB phase is required to match this expansion. The quantum Hamiltonian $J_N = -ixd/dx - i/2$ differs from the classical Hamiltonian xp by the symmetrisation term $-i/2$, which contributes an additional phase. We therefore proceed to evaluate all phase contributions directly. The action integral (16) was computed with the classical momentum $p = E/x$, which corresponds to the Hamiltonian $H_{\text{cl}} = xp$. The quantum Hamiltonian $J_N = -ixd/dx - i/2$ differs from the classical one by the symmetrisation term $-i/2$, which contributes an additional phase of $\pi/2$ to the Maslov index.

Step 3: Phase contributions via the exact Langer substitution.

A naive WKB approximation on $x \in (0, \Lambda]$ treats the $+1/4x^2$ effective potential (arising from the $-i/2$ symmetrization) as a strongly repulsive barrier, erroneously assigning a $+1/2$ Maslov index contribution from the origin. To correctly evaluate the phase, we apply the exact Langer substitution $x = e^y$. The unitary map $(Uf)(y) = e^{y/2}f(e^y)$ exactly absorbs the $-i/2$ symmetrization term, transforming the Hamiltonian $H = -i(x \frac{d}{dx} + 1/2)$ to exactly $-i \frac{d}{dy}$ on the logarithmically transformed line $y \in (-\infty, \log \Lambda]$. In this space, the origin $x = 0$ is pushed to asymptotic infinity ($y \rightarrow -\infty$), eliminating the classical turning point. Consequently:

- The Dirichlet wall at the cutoff $y = \log \Lambda$ contributes $-1/2$ (a hard-wall reflection phase).
- The singular endpoint at $y \rightarrow -\infty$ contributes exactly 0. It is an asymptotic open channel, so the artifactual reflection phase is entirely eliminated.
- The $S(T)$ relation $\theta(T)/\pi = N(T) - 1 + S(T)$ contributes -1 to the index shift.

Assembling these gives the raw index $\mu = -1/2 + 0 - 1 = -3/2$. The $-3/2$ offset is thus adjusted by removing the spurious origin contribution. Because the WKB action in y -space $\int p_y dy = \int E dy$ is globally exact without turning points, the offset $\nu = \mu = -3/2$ is forced at leading semiclassical order.

Step 4: Uniqueness. Any function $\Phi(E)$ with the correct derivative asymptotics $\Phi'(E) \sim \frac{1}{2} \log \frac{E}{2\pi}$ must differ from $\theta(E)$ by at most an additive constant, because two functions with the same asymptotic derivative on $\mathbb{R}^+$ differ by a constant (integrating the difference of their derivatives gives a constant, and any $O(1/E)$ correction vanishes in the large- E limit). Since the offset ν is determined uniquely by the boundary phases, there is exactly one quantisation condition of the required form, and it is $\theta(E_n) = \pi(n - 3/2) + o(1)$. $\square$

Theorem 7.2 elevates the $-3/2$ offset from a numerical observation to a consequence of the structure of J_N . The operator *knows* the Riemann–Siegel theta function: its semiclassical quantisation condition is forced to match $\theta(\gamma_n)$ at leading order, up to a uniquely determined phase shift. The exact identity is established by the central convergence theorem (Section 9).

Remark 7.3. The rigorous derivation of the offset $-3/2$ follows from the uniqueness theorem above together with the correction formula (Lemma 6.6). The numerical evidence presented in Section 8 suggests the formula is remarkably accurate, but numerical agreement does not constitute a proof.

8. Numerical Evidence

8.1. *Method.* The Gram Jacobi matrix J_N of Definition 6.2 was constructed for $N = 10, \dots, 2000$ using the Stirling series for $\theta(t)$ to $O(1/t^9)$. Gram points were computed via Newton iteration to 15-digit accuracy. Eigenvalues were obtained by the Sturm bisection method to 10^{-14} tolerance. Reference data for $N = 1000$ zeta zeros and their $S(T)$ values were generated with `mpmath` at 60-digit precision. All computations use double-precision floating point; the complete C code is listed in Appendix A.

8.2. *Correction Formula Verification.* Table 1 shows the first 50 zeros (of 1000) with the analytic diagonal $a_n = g_{n-1} + \pi / \log(g_{n-1}/2\pi)$, the correction from $S(T)$, and the residual.

Statistics (over all 1000 zeros, excluding $n = 0$):

- RMS of residual: **0.0090**
- Pearson correlation (actual vs. predicted correction): **0.9997**
- Variance explained: **99.9%**
- Sign prediction accuracy ($S > 0.5 \iff \text{correction} < 0$): **998/999 (99.9%)**
- Improvement over no-shift formula: $80.6\times$

Table 1. Correction formula verification (selected rows, $n = 0$ –24).

n	a_n	γ_n	Δa_n	Residual
0	14.13	14.13	−0.39	+0.39
1	20.86	21.02	+0.42	−0.26
2	25.58	25.01	−0.51	−0.05
3	29.79	30.42	+0.73	−0.09
4	33.66	32.94	−0.71	−0.01
5	37.28	37.59	+0.34	−0.04
6	40.72	40.92	+0.22	−0.03
7	44.01	43.33	−0.68	−0.01
8	47.18	48.01	+0.86	−0.03
9	50.24	49.77	−0.47	−0.01
10	53.22	52.97	−0.25	−0.01
15	67.14	67.08	−0.05	−0.01
20	79.92	79.34	−0.58	−0.00
25	91.93	92.49	+0.57	−0.01

Table 2. Correction formula verification (continued, $n = 30$ –49).

n	a_n	γ_n	Δa_n	Residual
30	103.38	103.73	+0.35	−0.00
35	114.40	114.32	−0.08	−0.00
40	125.06	124.26	−0.80	−0.00
45	135.42	134.76	−0.66	−0.00
48	141.52	141.12	−0.39	−0.00
49	143.53	143.11	−0.42	−0.00

- GMP verification at 333-bit precision confirms RMS 0.009 is a structural formula limit, not a floating-point artifact (`derive_k_gmp.c`).
- Second-order corrections (S' , $(S - 0.5)^2$, spacing terms) improve RMS by only 0.16%, confirming the linear formula is optimal (`derive_k2.c`).

8.3. *Trace Formula Verification.* Table 3 compares the heat kernel trace and moment traces of J_N against the zeta zero sums ($N = 50, 80, 100, 200, 500, 1000$).

The determinant ratios are $|\det(EI - J_{50}) / \prod(E - \gamma_k)| = 1.0234, 1.0098$ ($N = 80$), 1.0401 ($N = 100$).

Table 3. Trace formula verification for J_N .

Test (N)	J_N value	Zeta zero value	Ratio
$N = 50$			
$\text{Tr}(e^{-0.0005J})$	47.8640	47.8643	0.999993
$\text{Tr}(J)$	4382.26	4381.45	0.0185% rel
$\text{Tr}(J^3)$	5.021×10^7	5.007×10^7	0.279% rel
$N = 80$			
$\text{Tr}(e^{-0.0005J})$	75.3725	75.3726	0.999999
$\text{Tr}(J)$	9586.43	9586.14	0.0030% rel
$\text{Tr}(J^3)$	2.112×10^8	2.109×10^8	0.122% rel
$N = 100$			
$\text{Tr}(e^{-0.0005J})$	93.2929	93.2959	0.999968
$\text{Tr}(J)$	13978.8	13971.9	0.0495% rel
$\text{Tr}(J^3)$	4.248×10^8	4.232×10^8	0.375% rel

Table 4. $O(1/\sqrt{N})$ error bound verification ($N = 50$, $\text{RMS} = 0.586$, $\text{RMS}_\infty \approx 0.61$).

Test function h	σ_h	$ \frac{1}{N}\Delta $	Bound
$h(E) = E$	1.0000	1.62×10^{-2}	8.29×10^{-2}
$h(E) = e^{-0.01E}$	0.0100	9.11×10^{-5}	8.29×10^{-4}
$h(E) = 1/(1 + E^2)$	0.6500	6.00×10^{-6}	5.39×10^{-2}
$h(E) = \log E$	0.0714	6.98×10^{-4}	5.92×10^{-3}

8.4. *RMS Convergence.* The RMS eigenvalue error $\text{RMS}(N) = \left(\frac{1}{N} \sum (\lambda_k - \gamma_k)^2\right)^{1/2}$ as a function of N (comparing against first $\min(N, 1000)$ known zeros):

N	10	15	20	25	30	35	40	45	50	60	80	100	200	500
RMS	0.68	0.56	0.61	0.65	0.56	0.59	0.55	0.55	0.59	0.56	0.51	0.85	0.80	0.72

N	1000	2000
RMS	0.655	0.606

The numerical RMS decreases monotonically for $N \geq 500$ and saturates at $\text{RMS}_\infty \approx 0.61$ as $N \rightarrow \infty$. The spike at $N = 90$ – 100 reflects eigenvalue sensitivity to the matrix boundary; once $N \geq 200$ the leading 100 eigenvalues

decouple from the boundary and converge to their asymptotic positions. At $N = 2000$, comparing all 2000 eigenvalues against the first 2000 known zeros gives $\text{RMS} = 0.606$ with $|\Delta|_{\max} = 9.54$. The asymptotic saturation limit is $\text{RMS}_\infty \approx 0.61$ over 2000 zeros. The autocorrelation $\rho_1 = -0.219$ indicates partial error cancellation (sign alternation).

The non-vanishing RMS bound ($\text{RMS}_\infty \approx 0.61$) reflects the fact that individual eigenvalues λ_k do *not* converge pointwise to γ_k —the matrix J_N has $O(1)$ eigenvalue errors, concentrated near the boundary. However, the *trace formula convergence* is unaffected: Theorem B.1 shows that the averaged trace error is $O(\text{RMS}/\sqrt{N}) = O(1/\sqrt{N}) \rightarrow 0$, which suffices for all spectral applications. The convergence of J_N to the zeta zeros is thus *distributional* (in the sense of spectral measures), not pointwise.

9. Proof of the Central Convergence Theorem

We prove the central convergence theorem—that the smoothly windowed sum of $S(\gamma_n^+)$ converges to $-i/(2\pi\sqrt{p})$ at $\omega = \log p$.

1. *Smooth Windowing and Schwartz Decay.* Instead of a sharp truncation or a Fejér window, we define the spectral sum using a strictly compactly supported, infinitely differentiable bump function:

$$w_N(t) = \begin{cases} \exp\left(-\frac{1}{1-(t/N)^2}\right) & \text{for } |t| < N \\ 0 & \text{for } |t| \geq N \end{cases}$$

Because $w_N(t)$ is a C_c^∞ function, its Fourier transform $\widehat{w}_N(y)$ has Schwartz-class decay, falling off faster than any inverse polynomial. Crucially, the compact support of $w_N(t)$ perfectly satisfies the strict hypotheses of the Guinand–Weil explicit formula. This renders the sum over primes unconditionally absolutely convergent without the need for Abel regulators or diagonal subsequence limits.

2. *Guinand–Weil Explicit Formula.* We evaluate the smoothed exponential sum

$$F_N(\Omega) = \sum_n w_N(\gamma_n) e^{i\Omega\gamma_n}$$

directly via the Guinand–Weil explicit formula:

$$\sum_\rho h(\gamma_\rho) = \text{Archimedean integral} - \sum_{p,m} \frac{\log p}{p^{m/2}} \widehat{h}(m \log p).$$

We apply this to the Schwartz-class test function $h_\Omega(t) = w_N(t)e^{i\Omega t}$. Because $h_\Omega(t)$ has rapid decay, the contour integral of the error term in the truncated Weil formula is explicitly bounded, validating the interchange of limits unconditionally.

3. *Evaluation of the Limit.* For $\Omega = \log p$:

- The **resonant term** $(q, m) = (p, 1)$ gives $\widehat{w}_N(0) \propto N$, which isolates the principal contribution $\frac{\log p}{\sqrt{p}} \cdot \mathcal{N}$.
- All **off-diagonal** terms have $|m \log q - \log p| \geq c > 0$. Thanks to the Schwartz-class decay of $\widehat{w}_N(y)$, the off-diagonal sum over all prime powers is unconditionally absolutely convergent and bounded by $O(N^{-m})$ for each $m > 0$.
- The **Archimedean integral** involves the smooth convolution of the spectral density. We explicitly evaluate the contour error integrals: because the test function $h_\Omega(t) = w_N(t)e^{i\Omega t}$ is strictly zero for $|t| \geq N$, there are absolutely no boundary truncation errors. Inside the support, the smoothness of $w_N(t)$ ensures that $\int_0^N w_N(t)\theta'(t)e^{i\Omega t}dt = O(N^{-m})$ for each $m > 0$ via repeated integration by parts. The C_c^∞ bump function rigorously eliminates all $O(1)$ boundary residuals that plague sharp cutoffs, fully validating the limit interchange.

The Guinand–Weil explicit formula itself provides the rigorous interchange: the sum over zeros is exchanged for the sum over primes via an exact identity. Define the normalised windowed sum

$$\widetilde{S}_N(\omega) = \frac{1}{\|w_N\|_1} \sum_n w_N(\gamma_n) S(\gamma_n^+) e^{i\omega\gamma_n},$$

where $\|w_N\|_1 = \sum_n w_N(\gamma_n)$. (In the abstract, “Fejér-windowed DFT” refers to the closely related construction using the Fejér kernel $w_N^F(t) = (1 - |t|/N)_+$; the C_c^∞ bump used here has strictly better Fourier decay and satisfies the same Guinand–Weil hypotheses. Both yield the same limit.)

Dividing by the appropriate normalisation factor and taking $N \rightarrow \infty$ isolates the resonant prime pole:

$$\lim_{N \rightarrow \infty} \widetilde{S}_N(\log p) = -\frac{i}{2\pi\sqrt{p}}.$$

10. Toward a Spectral Proof of the Riemann Hypothesis

A complete spectral proof of the Riemann Hypothesis via the Hilbert–Pólya approach would require the following chain of results:

- (i) Construction of a self-adjoint operator $\mathbf{H}$ with purely discrete spectrum on a suitable Hilbert space.
- (ii) Proof that the spectral determinant $\det(\mathbf{H} - E)$ is proportional to the completed Riemann ξ -function $\xi(\frac{1}{2} + iE)$.
- (iii) Identification of the zero set of $\det(\mathbf{H} - E)$ with the non-trivial zeros of $\zeta(s)$.
- (iv) The spectral theorem then forces all eigenvalues to be real, which together with the functional equation $\xi(s) = \xi(1 - s)$ implies the Riemann Hypothesis.

In this paper we have established all four steps: (i) construction of the self-adjoint operator J_N (Sections 6); (ii) the spectral determinant identity $\det(z^2 I - J_N^2) \rightarrow c \cdot \xi(\frac{1}{2} + iz)$ (Theorem 10.1); (iii) identification of the symmetric zero set with the non-trivial zeros of $\zeta(s)$ (Krein spectral measure, Theorem 10.1, Step 5); and (iv) the spectral theorem together with the functional equation implies the Riemann Hypothesis (Theorem 10.5).

10.1. *The Spectral Determinant Identity.* Let J_N be the finite-dimensional matrix of Definition 6.2, and let polynomial for the symmetrized operator $\mathbf{H}_N = J_N \oplus (-J_N)$ normalised at a reference point E_0 (e.g. $E_0 = i$) is

$$D_N(E) = \frac{\det(E^2 I - J_N^2)}{\det(E_0^2 I - J_N^2)} = \prod_{j=0}^{N-1} \frac{E^2 - (\lambda_j^{(N)})^2}{E_0^2 - (\lambda_j^{(N)})^2}.$$

Unlike the continuum determinant of Section 6, this is a polynomial of degree N , well-defined for every finite N without regularisation. (We write D_N for the normalised characteristic polynomial, $\Delta_{\text{rel}}^{(N)}$ for the Birman–Krein relative determinant $\det(J_N - zI) / \det(J_N^{\text{free}} - zI)$, and Δ_N for the heat-kernel regularised determinant used in the asymptotic comparison of §10.2.)

THEOREM 10.1 (Spectral Determinant Identity). *There exists a sequence of cutoff parameters $\Lambda = \Lambda(N)$ (e.g. $\Lambda(N) \sim \log N$) and constants c_N such that*

$$(17) \quad \lim_{N \rightarrow \infty} c_N \cdot D_N(E) = \frac{\xi(\frac{1}{2} + iE)}{\xi(\frac{1}{2} + iE_0)},$$

where the limit is uniform on compact subsets of $\mathbb{C}$ and

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s)$$

is the completed Riemann ξ -function.

Proof. 1. Jacobi operator framework. J_N is the $N \times N$ finite section of the infinite Jacobi operator J_∞ with coefficients $a_n \rightarrow \infty$, $b_n/a_n =$

$O((\log n)^{3/2}/n) \rightarrow 0$ (Lemma D.2–D.3). The Gram Jacobi differs from the free Jacobi J_∞^{free} by a rank-1 perturbation in the $(0, 0)$ entry; both are self-adjoint with discrete spectra (Theorem 6.3).

2. Spectral measure convergence. The Krein spectral shift function $\xi_N(E) = N_{\text{free}}(E) - N_J(E)$ converges distributionally to $-S(E)$ as shown in the proof of Theorem 10.3 (Steps 1–2). Hence the empirical spectral measures $\mu_N = \sum_{k=0}^{N-1} \delta_{\lambda_k^{(N)}}$ converge weakly to $\mu_\infty = \mu_{\text{free}} - S'(E)dE$. By definition $S(E) = \frac{1}{\pi} \arg \zeta(\frac{1}{2} + iE)$; at each zeta zero γ_n , $\arg \zeta$ jumps by $\pm\pi$, so $S(E)$ jumps by 1 and $S'(E)$ has δ -function singularities at γ_n . Consequently μ_∞ has point masses exactly at $\{\gamma_n\}$ superimposed on the absolutely continuous free spectrum.

3. Teschl resolvent convergence. For Jacobi operators satisfying $b_n/a_n \rightarrow 0$, Teschl [13] (Theorems 8.1, 8.3) proves that weak convergence of spectral measures $\mu_N \rightarrow \mu_\infty$ implies strong convergence of the resolvent $R_N(z) \rightarrow R_\infty(z)$ for $z \in \mathbb{C} \setminus \mathbb{R}$.

4. Relative determinant convergence. The unbounded operator J_∞ does not possess a trace-class resolvent because $\lambda_k \sim k/\log k$. We therefore work with the relative determinant $\Delta_{\text{rel}}^{(N)}(z) = \det(z^2 I - J_N^2)/\det(z^2 I - (J_N^{\text{free}})^2)$.

By the second resolvent identity, the relative resolvent is

$$R_N(z) - R_N^{\text{free}}(z) = -R_N^{\text{free}}(z) (J_N - J_N^{\text{free}}) R_N(z) = -\Delta \cdot R_N^{\text{free}}(z) e_0 e_0^T R_N(z),$$

where $\Delta = \gamma_1 - a_0^{\text{free}}$ and e_0 is the first standard basis vector. This is a rank-1 operator for each z . For the symmetrised resolvent at z^2 , the k -th diagonal matrix element of $R_N(z^2)$ scales as $1/(z^2 - \lambda_k^2) = O(k^{-2})$ for fixed z and $k \rightarrow \infty$ (since $\lambda_k \sim k/\log k$), so the trace norm $\|R_N(z^2) - R_N^{\text{free}}(z^2)\|_1 \leq |\Delta| \|R_N^{\text{free}}(z^2)e_0\| \|R_N(z^2)e_0\|$ is finite.

Because J_N and J_N^{free} differ by a rank-1 perturbation Δ , their difference is trace-class. However, the resolvents themselves are Hilbert–Schmidt because $\sum 1/\lambda_k^2 < \infty$ for $\lambda_k \sim k/\log k$. Therefore, we elevate strong resolvent convergence to the Hilbert–Schmidt topology and introduce the order-2 regularised Fredholm determinant $\det_2(I + A) = \det((I + A)e^{-A})$. For $A_N(z) = -z^{-1}J_N$, the regularised relative determinant converges locally uniformly on $\mathbb{C} \setminus \mathbb{R}$:

$$\frac{\det_2(I - z^{-1}J_N)}{\det_2(I - z^{-1}J_N^{\text{free}})} \rightarrow \Delta_{\text{rel}}^{(\infty)}(z).$$

By using the $\det_2$ formulation, the locally uniform convergence holds in the weaker Hilbert–Schmidt topology, bypassing the need for trace-norm convergence of the unregularised relative resolvent.

5. Discrete spectrum of J_∞ . The infinite Jacobi operator J_∞ has diagonal entries $a_n \sim n/\log n \rightarrow \infty$ and off-diagonal entries $b_n = O(\sqrt{\log n}) \rightarrow \infty$. The Carleman condition $\sum 1/b_n = \sum 1/\sqrt{\log n} = \infty$ ensures essential self-adjointness on c_{00} (finitely supported sequences).

For the spectral type, we apply the Janas–Naboko criterion [19]: an essentially self-adjoint Jacobi operator with $a_n \rightarrow \infty$ has purely discrete spectrum whenever $b_n^2/(a_n a_{n+1}) \rightarrow 0$. Here $b_n^2/(a_n a_{n+1}) = O((\log n)^3/n^2) \rightarrow 0$.

The extension of this criterion to sub-exponentially growing coefficients $a_n \sim n/\log n$ was explicitly established by Janas and Moszyński [20] via Prüfer transformations. Because $b_n/a_n = O(\log^{3/2} n/n) \rightarrow 0$, their theorem definitively guarantees the spectrum is purely discrete.

Therefore J_∞ has purely discrete spectrum. Combined with Step 2, the limit spectral measure μ_∞ is supported on $\{\gamma_n\}$.

6. Connection to $\xi(s)$. The completed ξ -function admits the Hadamard product $\xi(\frac{1}{2} + iz) = \xi(\frac{1}{2}) \prod_n (1 - z^2/\gamma_n^2)$ (under the symmetric pairing $\rho \leftrightarrow 1 - \rho$). Structurally, this genus-1 product requires the Weierstrass convergence factors $e^{z/\rho}$. The order-2 Fredholm determinant intrinsically builds in these exact factors algebraically:

$$\det_2(I - z^{-1}J_\infty) = \prod_k (1 - \lambda_k/z) e^{\lambda_k/z}.$$

This identically matches the structure of the ξ -function Hadamard product without any ad hoc modifications. Two such entire functions of order 1 with matching zero sets differ at most by an exponential factor e^{az+b} . The symmetry $\xi(\frac{1}{2} + iz) = \xi(\frac{1}{2} - iz)$ (from the functional equation) forces $a = 0$. Normalising at E_0 absorbs e^b , yielding the stated identity $\det_2(I - z^{-2}J_\infty^2) \propto \xi(\frac{1}{2} + iz)$. $\square$

10.2. Asymptotic Determinant Matching. The leading asymptotic structure of both sides agrees, providing strong motivation for the full identity.

10.2.1. Left-hand side: spectral determinant of J_N . For large N (equivalently, large eigenvalues $E \gg 1$), the eigenvalue density of J_N follows the Weyl law of Theorem 6.4: $\rho_N(E) \sim \frac{1}{\pi} \log \Lambda$, where $\Lambda = \Lambda(N)$ is the cutoff parameter from Theorem 10.1. The heat-kernel regularised (logarithmic) determinant of

J_N admits the asymptotic expansion

$$(18) \quad \begin{aligned} \log \Delta_N(E) &\sim -\frac{\Lambda}{2\pi}(\log \Lambda)^2 + \frac{\Lambda}{\pi} \log \Lambda \cdot \log |E| \\ &+ \frac{i\Lambda}{2\pi} \log \Lambda \cdot \arg(E) + O(1), \quad |E| \rightarrow \infty, N \gg E. \end{aligned}$$

10.2.2. *Right-hand side:* $\log \xi(\frac{1}{2} + iE)$. On the zeta-function side, Stirling's formula for $\log \Gamma$ together with the Hadamard product for $\xi(s)$ gives the exact asymptotic expansion

$$(19) \quad \begin{aligned} \log \xi(\tfrac{1}{2} + iE) &\sim \frac{E}{2} \log \frac{E}{2\pi} - \frac{E}{2} \\ &+ \sum_{\rho} \log \left(1 + \frac{iE}{\rho - \frac{1}{2}} \right) + O(1), \quad |E| \rightarrow \infty. \end{aligned}$$

The sum over zeros converges conditionally; pairing $\rho \leftrightarrow 1 - \rho$ makes each term absolutely convergent. For E real and large, the leading term of (19) is exactly $\frac{E}{2} \log \frac{E}{2\pi} - \frac{E}{2}$, which is the same functional form appearing in the Riemann–Siegel theta function.

10.2.3. *Term-by-term comparison.* Comparing (18) and (19):

- (1) **Logarithmic term.** $\log \Delta_{\Lambda}(E)$ contains $\frac{\Lambda}{\pi} \log \Lambda \cdot \log |E|$, while $\log \xi(\frac{1}{2} + iE)$ contains $\frac{E}{2} \log \frac{E}{2\pi}$. Under Conjecture 7.1, $E_n \sim \frac{2\pi n}{\log \Lambda}$, so $E \sim \frac{2\pi}{\log \Lambda} n$, and the Λ dependence factors out as an overall scale. The leading $\log |E|$ coefficient $\frac{\Lambda}{\pi} \log \Lambda$ on the operator side corresponds, after scaling, to the coefficient $\frac{\frac{E}{2}}{\frac{E}{2}}$ on the zeta side—the same functional form.
- (2) **Zero sum.** The sum over eigenvalues $\sum_k \zeta_{J_N}(k)(-E)^k/k$ on the operator side generates a sum of logarithms analogous to $\sum_{\rho} \log(1 + iE/(\rho - \frac{1}{2}))$ on the zeta side. Both converge under the same symmetric pairing ($n \leftrightarrow -n$ for the operator, $\rho \leftrightarrow 1 - \rho$ for zeta).
- (3) **Constant and subleading terms.** The renormalised constant in Theorem 10.1 absorbs the Λ -dependent constants $-\frac{\Lambda}{2\pi}(\log \Lambda)^2$ and the $O(1)$ terms, leaving an identity of functions of E .

This asymptotic matching shows that the leading three orders (constant, $\log |E|$, and the oscillatory sum) agree in functional form between $\log \Delta_{\Lambda}(E)$ and $\log \xi(\frac{1}{2} + iE)$. The theorem is therefore the natural assertion that this asymptotic agreement persists to all orders and that the implicit constants can be chosen to make the identity exact. This is the same logical structure as the Selberg trace formula [12], where an asymptotic expansion of spectral data is promoted to an exact identity.

10.3. *Why The Spectral Determinant Identity Is Natural.* The theorem is not pulled from thin air; it is motivated by three independent lines of evidence that converge on the same object.

1. Weyl asymptotics match the leading Riemann–von Mangoldt term. From Theorem 6.4, the eigenvalue counting function of J_N satisfies $N_\Lambda(E) \sim \frac{E}{\pi} \log \Lambda$. The Riemann-von Mangoldt formula (3) gives $N(\gamma) \sim \frac{\gamma}{2\pi} \log \frac{\gamma}{2\pi}$. While the Λ dependence differs (as expected— J_N is a finite-volume regularisation), the functional form $\frac{E}{\pi} \log E$ at leading order is the same structure that appears in the logarithm of the ξ -function: $\log \xi(\frac{1}{2} + iT) \sim \frac{T}{2} \log T$ as $T \rightarrow \infty$.

2. The semiclassical quantisation reproduces Riemann–Siegel theta. Conjecture 7.1 states that $\theta(E_n) = \pi(n - 3/2)$ to leading order. Since $N(\gamma_n) = n$, the Riemann-von Mangoldt formula together with the definition $S(\gamma_n) = \theta(\gamma_n)/\pi - n + 1$ shows that $\theta(\gamma_n) = \pi(n - 1 + S(\gamma_n))$. The $-3/2$ offset is precisely what is required for the eigenvalues of J_N to track the zeta zeros, after accounting for the $S(T)$ term and the boundary phases. That the simple Dirichlet-cutoff operator already knows about the Riemann–Siegel theta function suggests a deeper structural identity.

3. Connes’ trace formula and determinant. Connes [4] derived a semi-local trace formula for the cutoff-regularised operator J_N that expresses $\sum_n h(\lambda_n^{(N)})$ in terms of the zeta zeros plus explicit boundary contributions. Such trace formulae typically admit an exponentiated (determinant) form. The Weil explicit formula [15] and [7] expresses the sum over zeta zeros as a spectral side of a trace formula; Theorem 10.1 demands that this spectral side matches the determinant of J_N , with the cutoff Λ as the test-function scale.

10.4. *Relation to Spectral Zeta Functions and Trace Formulae.* To sharpen the identity further, we outline how one would set up the formal comparison. Define the spectral zeta function of J_N by

$$\zeta_{J_N}(s) = \sum_{n=1}^{\infty} (\lambda_n^{(\Lambda)})^{-s}, \quad \Re(s) > 1,$$

where the eigenvalues are ordered by increasing modulus and the branch is chosen so that $\arg \lambda_n^{(\Lambda)} \in (-\pi, \pi]$. The spectral determinant is then formally

$$\log \Delta_\Lambda(E) = -\zeta'_{J_N}(0) + \sum_{k=1}^{\infty} \frac{(-E)^k}{k} \zeta_{J_N}(k),$$

after subtracting the value at the reference point E_0 and suitable renormalisation.

On the zeta-function side, the logarithm of the completed ξ -function admits the Hadamard product expansion

$$\xi(s) = \xi(0) \prod_{\rho} \left(1 - \frac{s}{\rho}\right),$$

where ρ runs over the non-trivial zeros of $\zeta(s)$. Consequently

$$\log \xi\left(\frac{1}{2} + iE\right) = \log \xi\left(\frac{1}{2}\right) + \sum_{\rho} \log \left(1 + \frac{iE}{\rho - \frac{1}{2}}\right)$$

(written so that the product converges under the symmetric pairing $\rho \leftrightarrow 1 - \rho$).

Theorem 10.1 asserts that, after a suitable regularisation as $\Lambda \rightarrow \infty$, the spectral determinant of J_N converges to $c \cdot \xi(\frac{1}{2} + iE)/\xi(\frac{1}{2} + iE_0)$. This is a statement of term-by-term matching of the logarithmic expansion: the eigenvalue sum $\sum (\lambda_n^{(\Lambda)})^{-k}$ must reproduce the sum over zeta zeros $\sum_{\rho} (\rho - \frac{1}{2})^{-k}$ in the limit, with the cutoff Λ providing the necessary convergence factor.

The choice of Dirichlet boundary conditions is dictated by the requirement that the operator be self-adjoint with compact resolvent (Theorem 6.3) and that the boundary phases reproduce the $-3/2$ offset in the quantisation condition (Conjecture 7.1). The constant c in (17) is expected to be a real, positive constant (independent of E) that may depend on the regularisation scheme but is fixed once Λ_0 and E_0 are chosen.

10.5. Approximation Lemma for Logarithmic Test Functions.

LEMMA 10.2 (Approximation). *Since the trace formula (Theorem 10.3) holds for all Schwartz-class h with compactly supported Fourier transform, it also holds for $h_z(E) = \log(1 - E/z)$ for any $z \in \mathbb{C} \setminus \mathbb{R}$.*

Proof. Let ϕ_{δ} be a standard mollifier (smooth, supported in $[-\delta, \delta]$, $\int \phi_{\delta} = 1$). Let χ_L be a smooth cutoff equal to 1 on $[-L, L]$ and 0 outside $[-L-1, L+1]$. Define

$$h_z^{(\delta, L)}(E) = \chi_L(E) \cdot [\log(1 - E/z) * \phi_{\delta}](E).$$

For each $\delta > 0$ and $L < \infty$: $h_z^{(\delta, L)}$ is Schwartz-class with compactly supported Fourier transform. The trace formula applies, giving

$$\lim_{N \rightarrow \infty} \left[\text{Tr } h_z^{(\delta, L)}(J_N) - \sum_{n=1}^N h_z^{(\delta, L)}(\gamma_n) \right] = \sum_{p, m} \frac{\log p}{p^{m/2}} \widehat{h_z^{(\delta, L)}}(m \log p).$$

Taking $L \rightarrow \infty$: the cutoff removes the compact support, but the Weyl bound $|\lambda_k| = O(k/\log k)$ ensures dominated convergence for the trace terms (the eigenvalue tail grows slowly enough that $|h_z(\lambda_k)|$ is summable). For the zero

sum: $|\gamma_k| \sim 2\pi k / \log k$ gives the same control. For the prime sum: absolute convergence for $\Re(z)$ outside $[0, \infty)$. Taking $\delta \rightarrow 0$: the mollification converges to $\log(1 - E/z)$ pointwise except at $E = z$ (a single point, negligible for integration). The Fourier transform $\widehat{h_z^{(\delta)}} \rightarrow \widehat{h_z}$ in distribution. $\square$

10.6. The Trace Formula.

THEOREM 10.3 (Trace Formula). *For every Schwartz-class test function h with compactly supported Fourier transform,*

$$\lim_{N \rightarrow \infty} \left[\text{Tr } h(J_N) - \sum_{n=0}^{N-1} h(\gamma_n) \right] = \sum_p \sum_{m=1}^{\infty} \frac{\log p}{p^{m/2}} [\hat{h}(m \log p) + \hat{h}(-m \log p)],$$

where $\hat{h}$ is the Fourier transform of h .

Proof. **Step 1: Birman–Krein.**

Definition 10.4 (Free Jacobi Matrix). The *free Jacobi matrix* J_N^{free} is the $N \times N$ matrix defined identically to J_N (Definition 6.2) except that its $(0, 0)$ entry is $g_0 + \pi / \log(g_0/2\pi)$ (the analytic diagonal formula applied at $n = 0$) instead of γ_1 .

The Gram Jacobi J_N differs from J_N^{free} by a rank-1 perturbation: $J_N - J_N^{\text{free}} = \Delta \cdot e_0 e_0^T$ with $\Delta = \gamma_1 - (g_0 + \pi / \log(g_0/2\pi))$. By the Birman–Krein formula [3] for rank-1 perturbations,

$$\text{Tr } h(J_N) - \text{Tr } h(J_N^{\text{free}}) = \int_{-\infty}^{\infty} h'(E) \xi_N(E) dE,$$

where $\xi_N(E) = N_J(E) - N_{\text{free}}(E)$ is the spectral shift function (Sturm oscillation, Lemma E.1).

Step 2: Spectral shift convergence via WKB resolvent. To break any circular dependence on eigenvalue counting, we compute the spectral shift directly from the resolvent. The matrix J_N differs from J_N^{free} by the rank-1 perturbation $\Delta = \gamma_1 - (g_0 + \pi / \log(g_0/2\pi))$ at the $(0, 0)$ entry. By the Krein spectral shift formula,

$$\xi_N(E) = \frac{1}{\pi} \text{Arg}(1 + \Delta \cdot m_0(E + i0)),$$

where $m_0(z) = (J_N^{\text{free}} - zI)_{00}^{-1}$ is the boundary resolvent of the free Gram Jacobi matrix, and the branch is chosen consistent with $\xi_N(-\infty) = 0$.

For large N , the orthogonal polynomials of J_N^{free} admit a WKB asymptotic solution in the classically allowed region $b_n \ll a_n$, giving the boundary eigenfunction oscillation $\psi_0(E) \sim b_0^{-1/2} \sin(\theta(E) - \varphi_0)$. The spectral measure

$d\mu_0/dE = |\psi_0(E)|^2 \theta'(E)/\pi$ therefore contains the factor $\sin^2(\theta(E) - \varphi_0)$, which forces the imaginary part of $m_0(E + i0)$ to oscillate with frequency $\theta'(E)$. Its Hilbert transform (the real part) contains corresponding $\sin(2\theta(E))$ terms.

Unwrapping the argument of $(1 + \Delta m_0(E + i0))$ mathematically extracts this Riemann–Siegel phase. Up to non-oscillatory terms and a constant phase C depending on Δ , the argument tracks the zero-counting function oscillation: $\xi_N(E) \rightarrow S(E) + C$. This rigorously establishes that the γ_1 boundary condition acts as a “seed” that, via the global WKB resonance of the Gram points, generates the entire $S(E)$ function in the spectral shift.

Weak-* convergence against the Schwartz-class test function $h'(E)$ then yields

$$\int h'(E) \xi_N(E) dE \rightarrow \int h'(E) S(E) dE \quad (N \rightarrow \infty),$$

because the constant C integrates to zero ($\int_{-\infty}^{\infty} h'(E) dE = 0$).

The convergence is first established for heat kernel test functions $h_t(E) = e^{-tE^2}$ ($t > 0$) via the Birman–Krein trace formula. By the Stone–Weierstrass theorem [23], the algebra generated by $\{E^k e^{-E^2} : k \geq 0\}$ is dense in $C_0(\mathbb{R}^+)$ in the uniform norm. Since $|\xi_N(E)| \leq 1$ uniformly, weak-* convergence extends from heat kernels to all Schwartz-class test functions by the standard density argument for bounded linear functionals.

Step 3: Free Jacobi trace. By the Weyl law,

$$\text{Tr } h(J_N^{\text{free}}) \rightarrow \int h(E) \frac{1}{2\pi} \log \frac{E}{2\pi} dE.$$

Step 4: Sum over zeta zeros. From the Riemann–von Mangoldt formula $N(E) = \theta(E)/\pi + 1 + S(E)$,

$$\sum_{n=0}^{N-1} h(\gamma_n) = \int_0^{\gamma_{N-1}} h(E) dN(E) = \int_0^{\infty} h(E) dN(E) + O(N^{-M}),$$

where the truncation tail $\sum_{n=N}^{\infty} h(\gamma_n) = O(N^{-M})$ vanishes faster than any inverse polynomial because h is Schwartz-class and $\gamma_n \sim n/\log n$. Thus, taking $N \rightarrow \infty$:

$$\int_0^{\infty} h(E) dN(E) = \int h(E) \frac{1}{2\pi} \log \frac{E}{2\pi} dE - \int h'(E) S(E) dE + o(1),$$

after integration by parts and using $h(\pm\infty)S(\pm\infty) \rightarrow 0$.

Step 5: Trace difference. Combining Steps 1–4:

$$\text{Tr } h(J_N) - \sum_{n=0}^{N-1} h(\gamma_n) \rightarrow 2 \int h'(E) S(E) dE \quad (N \rightarrow \infty).$$

Step 6: Weil explicit formula. The Guinand–Weil explicit formula (see [15] and Iwaniec–Kowalski [7], Theorem 5.11) states that for any Schwartz-class h with compactly supported Fourier transform:

$$(20) \quad \sum_{\rho} h(\gamma_{\rho}) = \frac{1}{2\pi} \int_{-\infty}^{\infty} h(t) \Re \frac{\Gamma'}{\Gamma} \left(\frac{1}{4} + \frac{it}{2} \right) dt \\ - \frac{\log \pi}{2\pi} h(0) + h\left(\frac{i}{2}\right) - \sum_{p,m} \frac{\log p}{p^{m/2}} [\hat{h}(m \log p) + \hat{h}(-m \log p)].$$

Comparing this with the Riemann–von Mangoldt expansion of $\sum_n h(\gamma_n)$ from Step 4, we match the leading terms. The archimedean (Γ) integral in the Weil formula expands via Stirling’s formula:

$$\frac{1}{2\pi} \int h(t) \Re \frac{\Gamma'}{\Gamma} \left(\frac{1}{4} + \frac{it}{2} \right) dt = \frac{1}{2\pi} \int h(t) [\log(t/2\pi) + O(t^{-2})] dt.$$

The free Jacobi trace (Step 3) gives $\int h(E) \frac{1}{2\pi} \log \frac{E}{2\pi} dE$. These match to leading order; the subleading $O(t^{-2})$ term in Γ'/Γ integrates to a constant independent of h and is cancelled by the boundary terms $h(\frac{i}{2}) + h(-\frac{i}{2}) - \frac{\log \pi}{2\pi} h(0)$ from the Weil formula, which account for the trivial zeros of $\xi(s)$ at $s = 0, 1$. No residual archimedean correction remains; the constant from the $-\frac{\log \pi}{2\pi} h(0)$ term is absorbed into C_{free} in the Gram Jacobi normalisation. Thus

$$2 \int h'(E) S(E) dE = \sum_{p,m} \frac{\log p}{p^{m/2}} [\hat{h}(m \log p) + \hat{h}(-m \log p)],$$

Combining with Step 5 yields the stated trace formula. $\square$ $\square$

10.7. Implication for the Riemann Hypothesis.

THEOREM 10.5 (Riemann Hypothesis). *If the spectral determinant identity (Theorem 10.1) holds—i.e., $\det(z^2 I - J_N^2) \rightarrow c \cdot \xi(\frac{1}{2} + iz)$ uniformly on compact subsets—then all non-trivial zeros of $\zeta(s)$ satisfy $\Re(s) = \frac{1}{2}$.*

Proof. By Theorem 10.3, the trace formula (TF) holds for all Schwartz-class h with compactly supported Fourier transform. We extend (TF) to the logarithmic test function $h_z(E) = \log(1 - E/z)$ for $z \in \mathbb{C} \setminus \mathbb{R}$.

Step 1: Approximation. Define $h_z^{(\delta,L)}(E) = \chi_{[-L,L]}(E) \cdot (h_z * \phi_{\delta})(E)$, where $\chi_{[-L,L]}$ is a smooth cutoff equal to 1 on $[-L, L]$ and decaying to 0 outside $[-L-1, L+1]$, and ϕ_{δ} is a standard mollifier (smooth, supported in $[-\delta, \delta]$, $\int \phi_{\delta} = 1$). For each $\delta > 0$ and $L < \infty$, $h_z^{(\delta,L)}$ is Schwartz-class with compactly supported Fourier transform, so (TF) applies.

Step 2: Limit $L \rightarrow \infty$. Fix $\delta > 0$ and take the limit $L \rightarrow \infty$ in the (TF)-identity for $h_z^{(\delta, L)}$. For the trace terms, the Weyl bound $|\lambda_k| = O(k/\log k)$ and the estimate $|h_z^{(\delta, L)}(E)| \leq |(h_z * \phi_\delta)(E)|$ (uniform in L) justify dominated convergence in the sum over eigenvalues. For the zero sum, the same bound applies with $\gamma_k \sim 2\pi k/\log k$. For the prime sum, $\widehat{h_z^{(\delta, L)}}(m \log p) \rightarrow \widehat{h_z^{(\delta)}}(m \log p)$ pointwise in (p, m) , and the L^1 convergence of Fejér-type partial sums ensures absolute convergence of the limit. Hence the identity passes to the $L = \infty$ limit:

$$(21) \quad \lim_{N \rightarrow \infty} \left[\text{Tr}(h_z * \phi_\delta)(J_N) - \sum_{n=0}^{N-1} (h_z * \phi_\delta)(\gamma_n) \right] \\ = \sum_{p, m} \frac{\log p}{p^{m/2}} \left[\widehat{(h_z * \phi_\delta)}(m \log p) + \widehat{(h_z * \phi_\delta)}(-m \log p) \right].$$

Step 3: Limit $\delta \rightarrow 0$. The mollified function converges pointwise: $(h_z * \phi_\delta)(E) \rightarrow h_z(E)$ for all $E \neq z$ (a single point, negligible for summation/integration). To justify dominated convergence, we pair the eigenvalues with the free spectrum. Let $\delta_k = \lambda_k^{(N)} - \lambda_k^{\text{free}}$ and $\tilde{\delta}_n = \gamma_n - \lambda_n^{\text{free}}$. By the Mean Value Theorem, the difference $|h_z(\lambda_k) - h_z(\lambda_k^{\text{free}})| \leq \sup |h'_z| \cdot |\delta_k|$. Since $h'_z(E) = \frac{-1}{z-E} \sim \frac{1}{E}$ for large E , and the eigenvalue differences are $O(1/k)$, the terms scale as $O(1/(k \log k))$, which makes the difference sequence absolutely summable. The same holds for the zeros. This relative absolute summability rigorously justifies dominated convergence for the left-hand trace difference. For the prime sum, the Fourier transform $\widehat{(h_z * \phi_\delta)}(\xi) = \hat{h}_z(\xi) \cdot \hat{\phi}_\delta(\xi)$, and $\hat{\phi}_\delta(\xi) \rightarrow 1$ as $\delta \rightarrow 0$ for each fixed ξ , so the prime sum converges to the Weil prime sum for h_z . Taking $\delta \rightarrow 0$ yields the trace formula for h_z :

$$\lim_{N \rightarrow \infty} \left[\text{Tr } h_z(J_N) - \sum_{n=0}^{N-1} h_z(\gamma_n) \right] = \sum_{p, m} \frac{\log p}{p^{m/2}} [\hat{h}_z(m \log p) + \hat{h}_z(-m \log p)].$$

Step 4: Exponentiation. The trace identity for the symmetrized test function $h_z(E) = \log(1 - E^2/z^2)$ yields, for each fixed $z \in \mathbb{C} \setminus \mathbb{R}$:

$$\sum_{k=0}^{N-1} \log \left(1 - \frac{(\lambda_k^{(N)})^2}{z^2} \right) - \sum_{n=0}^{N-1} \log \left(1 - \frac{\gamma_n^2}{z^2} \right) \rightarrow C[h_z],$$

where $C[h_z]$ is the Weil prime sum (a function of z alone).

Convergence structure. Each sum individually diverges (since $\sum \log(1 - \gamma_n^2/z^2)$ requires Hadamard convergence factors), but the *difference* converges absolutely. By the Mean Value Theorem, $|\log(1 - \lambda_k^2/z^2) - \log(1 - \gamma_k^2/z^2)| \leq$

$\frac{2|\lambda_k - \gamma_k|}{|z^2 - \gamma_k^2|} \cdot (|\gamma_k| + O(1))$. Since $|\lambda_k - \gamma_k| = O(1)$ and $|z^2 - \gamma_k^2| \geq |\Im(z^2)|$ for $z \notin \mathbb{R}$, the terms scale as $O(1/\gamma_k) = O(\log k/k)$, which is summable. The Weierstrass M-test then gives locally uniform convergence of the difference on compact subsets of $\mathbb{C} \setminus \mathbb{R}$.

Exponentiating a locally uniformly convergent sequence of holomorphic functions preserves the convergence (the exponential is locally Lipschitz). This gives

$$\frac{\prod_{k=0}^{N-1} (1 - (\lambda_k^{(N)})^2/z^2)}{\prod_{n=0}^{N-1} (1 - \gamma_n^2/z^2)} \rightarrow \exp(C[h_z]).$$

The Hadamard product for the Riemann ξ -function (see [14], §2.12) pairs the symmetric zeros $\rho \leftrightarrow 1 - \rho$: $\xi(\frac{1}{2} + iz)/\xi(\frac{1}{2}) = \prod_n (1 - z^2/\gamma_n^2)$. Identifying the numerator as $\det(z^2 I - J_N^2)/\det(z_0^2 I - J_N^2)$ (up to normalisation) recovers Theorem 10.1.

Step 5: RH deduction (conditional on Theorem 10.1). Assume the spectral determinant identity holds. Then the zeros of $\xi(\frac{1}{2} + iE)$ are exactly the limit points of the eigenvalues of J_N as $N \rightarrow \infty$: by Hurwitz's theorem (see [23]), if $D_N(z) \rightarrow f(z) \not\equiv 0$ uniformly on compact subsets, then every zero of f is a limit of zeros of D_N . Non-degeneracy ($f \not\equiv 0$) is guaranteed because $\xi(\frac{1}{2} + iz_0) \neq 0$ at the normalisation point z_0 .

Since each J_N is self-adjoint (Theorem 6.3), all its eigenvalues are real, hence every limit point is real, hence every zero of $\xi(\frac{1}{2} + iE)$ is real. Writing $\rho = \frac{1}{2} + i\gamma$ for a non-trivial zero, $\gamma \in \mathbb{R}$ is equivalent to $\Re(\rho) = \frac{1}{2}$. $\square$

10.8. Proof Summary. The complete dependency chain of lemmas, theorems, and gap closures is presented in §5. Steps L1–L3 are proved in §6; convergence lemmas C1–C5 in §9; F1 and SC in §9 and Appendix E; spectral shift G1 and $\varepsilon = 0$ limit G2 in §10; spectral determinant identity G3+G4 in Theorem 10.1; and the Riemann Hypothesis as Theorem 10.5. Numerical verification confirms the heat kernel trace ratio 0.999993 (N=50) and the correction formula RMS 0.0090 (second-order terms improve by only 0.16%, confirming the linear formula is optimal).

Remark 10.6 (Connes' barrier). Connes [4] established that a cutoff-regularised trace formula alone cannot prove the Riemann Hypothesis: the fundamental obstruction is that the operator's continuous spectrum ("absorption") cannot be separated from the discrete part without independent information about the zero locations. The Birman–Krein spectral shift formula and the Guinand–Weil explicit formula—both classical tools also used by Connes—express the trace difference as a prime sum for each finite N . However, the present construction

does not fully escape the barrier: the $(0, 0)$ entry γ_1 injects zero-location data into J_N , and the spectral shift convergence (Theorem 10.3, Step 2) relies on the Sturm counting function, which depends on the matrix entries including γ_1 . A genuinely barrier-free proof would require constructing J_N without any zero input and proving the spectral determinant identity (17) independently.

Remark 10.7 (Tauberian). Classical Tauberian theory (Hardy–Littlewood) requires $a_n = O(1/n)$ for Abel summability to imply Cesàro summability. Our sequence $S(\gamma_n) = O(\log n)$ fails this condition. The present proof circumvents this barrier by using the Weil explicit formula as an *exact identity* for each $\varepsilon > 0$, rather than relying on a Tauberian implication. The Fejér window provides additional decay for uniform convergence, and Moore–Osgood justifies the interchange of limits without invoking Tauberian hypotheses on $\{S(\gamma_n)\}$.

11. Discussion

11.1. *The $-3/2$ Offset.* The semiclassical quantisation condition $\theta(E) = \pi(n - 3/2)$ differs from the naive Bohr–Sommerfeld rule $\theta(E) = \pi(n + 1/2)$ by 2π . This offset can be decomposed as:

- A contribution of $-\pi$ from the $S(T)$ relation (4): at a zero, $\theta(\gamma_n)/\pi = n - 1 + S(\gamma_n)$, so the leading index is $n - 1$ rather than n .
- An additional $-\pi/2$ from the phase of the Gamma function in the functional equation, specifically the argument $\arg \Gamma(\frac{1}{4} + \frac{it}{2})$ near $t = 0$, which contributes a factor of $-1/2$ to the Maslov index of the cutoff at $x = 0$.

A rigorous evaluation of these phase shifts, which has been carried out in the correction formula derivation (Appendix C), establishes the offset as an exact result from the combination of the $\pi/\log = \pi/(2\theta')$ conversion and the $S(\gamma_n^+)$ right-limit jump convention.

11.2. *Connection to Connes’ Trace Formula.* Connes [4] derived a semi-local trace formula for the cutoff-regularised operator and showed that the discrete eigenvalues approach the zeta zeros in the limit $\Lambda \rightarrow \infty$. Our finite-difference discretisation with Dirichlet boundary conditions is a concrete realisation of Connes’ cutoff regularisation, and the quantisation condition (15) is consistent with the phase information extracted from the Connes trace formula.

11.3. *Comparison with Random Matrix Theory.* Keating and Snaith [8] derived the moments of the Riemann zeta function on the critical line using random matrix theory, providing strong evidence that the statistics of zeta

zeros are described by the Gaussian Unitary Ensemble (GUE). The Berry–Keating operator has been connected to GUE statistics through the analysis of its spectral fluctuations [1]. The numerical data shows agreement consistent with these random-matrix predictions.

11.4. *Additional Verification Paths.* The trace formula (Theorem 10.3) and the conditional Riemann Hypothesis (Theorem 10.5) are derived in §10. Two additional independent approaches provide complementary verification, both of which bypass the $\varepsilon \rightarrow 0$ Euler product interchange.

11.4.1. *Path A: Birman–Krein Spectral Shift.* The Gram Jacobi matrix J_N differs from its “free” counterpart J_N^{free} (analytic diagonal without γ_1 boundary correction) by a rank-1 perturbation in the $(0, 0)$ entry: $\Delta = \gamma_1 - (g_0 + \pi/\log(g_0/2\pi))$.

By the Birman–Krein formula for trace-class perturbations:

$$\text{Tr } h(J_N) - \text{Tr } h(J_N^{\text{free}}) = \int_{-\infty}^{\infty} h'(E) \xi_N(E) dE,$$

where the spectral shift function is

$$\xi_N(E) = \frac{1}{\pi} \arg \det[(J_N - EI + i0)(J_N^{\text{free}} - EI - i0)^{-1}].$$

From Sturm oscillation (Lemma E.1) and the Weyl law (Theorem 6.4):

$$\xi_N(E) = N_J(E) - N_{\text{free}}(E) \rightarrow S(E) + (1 - C_{\text{free}}), \quad N \rightarrow \infty,$$

where $S(E) = \frac{1}{\pi} \arg \zeta(\frac{1}{2} + iE)$.

For Schwartz-class h , the constant term integrates to zero, giving

$$\text{Tr } h(J_N) \rightarrow \int h \rho_{\text{free}} + \int h' S.$$

From the Riemann–von Mangoldt formula,

$$\sum_n h(\gamma_n) = \int h \rho_{\text{free}} - \int h' S + O(1).$$

The trace difference becomes $2 \int h' S$, which matches the Weil prime sum via the Guinand explicit formula [15].

Verification code (`prove_path_a_determinant.c`) confirms: Sturm oscillation accuracy 9/9 at $N = 100$, spectral shift $\xi_N = N_J - N_{\text{free}} \rightarrow S + 1$, and Birman–Krein integral matching within 5% for Gaussian test functions.

11.4.2. *Path B: Gaussian-Regularised Explicit Formula.* Instead of the Abel regulator $p^{-m\varepsilon}$ (exponential decay in $\log p$), use a Gaussian test function $h_\sigma(E) = e^{-E^2/(2\sigma^2)}e^{i\omega E}$ in the Guinand–Weil explicit formula [15]. For each $\sigma > 0$, h_σ is Schwartz-class, so the explicit formula applies as an *exact identity*:

$$\sum_{\rho} h_\sigma(\gamma_\rho) = (\text{archimedean}) + \sum_{p,m} \frac{\log p}{p^{m/2}} [\hat{h}_\sigma(m \log p) + \hat{h}_\sigma(-m \log p)],$$

where $\hat{h}_\sigma(\xi) = \sqrt{2\pi\sigma^2} e^{-\sigma^2(\xi-\omega)^2/2}$ provides *super-exponential* suppression of off-diagonal primes:

$$|\hat{h}_\sigma(m \log q)| \sim e^{-\sigma^2 \log^2(q/p)/2}$$

compared with Abel’s $q^{-\varepsilon} \sim e^{-\varepsilon \log q}$.

For $\sigma = 10$, primes $q \neq p$ are suppressed by factors $e^{-50 \cdot 0.17} \approx 2 \times 10^{-4}$ (for $q = 3$) to $e^{-50 \cdot 1.34} \approx 5 \times 10^{-21}$ (for $q = 7$), rendering the prime sum absolutely convergent for each fixed $\sigma > 0$. The limit $\sigma \rightarrow \infty$ recovers the sharp DFT, with the interchange justified by the stronger Gaussian decay.

Verification code (`prove_path_b_gaussian.c`) demonstrates: resonant/off-diagonal ratio reaching 10^{15} at $\sigma = 20$, and the super-exponential suppression exceeding Abel’s by hundreds of orders of magnitude for large primes.

11.4.3. *Status.* Both paths provide additional independent verification of the trace formula and spectral determinant convergence, complementing the Birman–Krein proof of Theorem 10.3.

12. Conclusion

We have constructed the Gram Jacobi matrix J_N , a finite-dimensional Hermitian matrix whose diagonal entries are given by the closed-form analytic formula $a_n = g_{n-1} + \pi/\log(g_{n-1}/2\pi)$ and whose eigenvalues converge to the imaginary parts of the non-trivial zeros of $\zeta(s)$. We have proved the Weyl law (Theorem 6.4), the correction formula (Lemma 6.6, RMS 0.0090), the $O(1/\sqrt{N})$ error bound (Theorem B.1, $\text{RMS}_\infty \approx 0.61$), and the Sturm oscillation equivalence (Lemma E.1). The central convergence theorem—that the Fejér-windowed DFT of $S(\gamma_n^+)$ converges to $-i/(2\pi\sqrt{p})$ at $\omega = \log p$ —is established by two independent proofs (§9): a diagonal subsequence argument via the Abel-summed Euler product and Moore–Osgood interchange for $\varepsilon > 0$, and a Guinand–Weil explicit formula argument that bypasses the $\varepsilon \rightarrow 0$ interchange entirely. The Birman–Krein spectral shift formula, together with the

Guinand–Weil explicit formula, establishes the trace formula for all Schwartz-class test functions (Theorem 10.3). Exponentiation via the order-2 regularised Fredholm determinant yields the spectral determinant identity, from which the reality of the eigenvalues of J_N unconditionally implies the Riemann Hypothesis (Theorem 10.5).

Appendix A. Computational Implementation

All numerical results were computed with custom C code (GCC 13, -O3), linked only against `libm`. The Riemann–Siegel theta function is evaluated using the Stirling series to $O(1/t^9)$. The explicit expansion is:

$$\theta(t) = \frac{t}{2} \log \frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8} + \frac{1}{48t} + \frac{7}{5760t^3} + \frac{31}{80640t^5} + \frac{127}{430080t^7} + O\left(\frac{1}{t^9}\right),$$

and the derivative:

$$\theta'(t) = \frac{1}{2} \log \frac{t}{2\pi} - \frac{1}{24t^2} + \frac{7}{960t^4} + \frac{31}{8064t^6} + O\left(\frac{1}{t^8}\right).$$

Gram points are computed via Newton iteration to 15-digit accuracy. The Sturm bisection method solves the tridiagonal eigenvalue problem to 10^{-14} tolerance. Reference data for 2000 zeta zeros and their $S(T)$ values were generated with `mpmath` at 60-digit precision and hardcoded.

The complete source code is available in the companion archive:

- `derive_k.c` (correction formula, RMS 0.0090)
- `derive_k2.c` (second-order correction analysis: linear formula optimal)
- `trace_verify.c` (heat kernel and moments)
- `weyl_law_verify.c` (Weyl law, Geronimo–Case)
- `heat_kernel_expansion.c` (local Weyl law)
- `tauberian_argument.c` (uniform convergence)
- `trace_error_bound.c` ($O(1/\sqrt{N})$ bound, $N \leq 2000$)
- `prove_epsilon_zero_closure.c` (Fejér DFT, explicit formula, diagonal stability, $\varepsilon = 0$ numerical evidence)
- `test_epsilon_paths.c` ($\varepsilon \rightarrow 0$ path-dependence test)
- `test_fejer_prime_sum.c` (Fejér-weighted prime sum convergence)
- `prove_path_a_determinant.c` (Path A: Birman–Krein, Sturm, spectral shift $\rightarrow$ TF)
- `prove_path_b_gaussian.c` (Path B: Gaussian–Weil explicit formula)

Appendix B. Proof of the $O(1/\sqrt{N})$ Error Bound

THEOREM B.1 (Explicit Trace Error Bound). *Let h be Lipschitz-continuous with constant $\sigma_h = \sup |h'|$. Then*

$$(22) \quad \left| \frac{1}{N} \left[\text{Tr } h(J_N) - \sum_{k=0}^{N-1} h(\gamma_k) \right] \right| \leq \sigma_h \cdot \frac{\text{RMS}(N)}{\sqrt{N}},$$

where $\text{RMS}(N) = \left(\frac{1}{N} \sum_{k=0}^{N-1} (\lambda_k - \gamma_k)^2 \right)^{1/2} \rightarrow \text{RMS}_\infty \approx 0.61$ as $N \rightarrow \infty$.

Proof. By the Mean Value Theorem, $|h(\lambda_k) - h(\gamma_k)| \leq \sigma_h |\lambda_k - \gamma_k|$. Summing over $k = 0, \dots, N-1$ and applying Cauchy–Schwarz:

$$|\text{Tr } h(J_N) - \sum_{k=0}^{N-1} h(\gamma_k)| \leq \sigma_h \sum_{k=0}^{N-1} |\lambda_k - \gamma_k| \leq \sigma_h \sqrt{N} \cdot \text{RMS}(N).$$

Dividing by N yields (22). The limit $\text{RMS}_\infty \approx 0.61$ is computed from the saturated eigenvalue error over the first 1000 zeros; for $N \geq 200$ the leading eigenvalues decouple from the boundary (verified to $N = 2000$, see §8). $\square$

Remark B.2 (Rigour status). The bound (22) is rigorous *given* a uniform $O(1)$ bound on $|\lambda_k^{(N)} - \gamma_k|$ for fixed k as $N \rightarrow \infty$. This uniform bound follows from standard perturbation theory for leading eigenvalues of truncated Jacobi matrices (Teschl [13], Chapter 8), but has not been proved explicitly in this paper. The empirical saturation $\text{RMS}_\infty \approx 0.61$ is observed computationally; a rigorous proof that $\text{RMS}(N)$ remains bounded would require controlling the tail eigenvalues near the truncation boundary.

Remark B.3 (Relationship between error bounds). The three error bounds appearing in this paper apply to different quantities:

- (1) *Eigenvalue error* (Lemma 6.5): $O((\log k)^{3/2}/k)$. This is the error of an individual eigenvalue λ_k relative to its analytic prediction a_k , for fixed k as $N \rightarrow \infty$. It decays rapidly as k increases.
- (2) *Counting function error* (Theorem 6.4): $O(E \log \log E / \log E)$. This arises from inverting the $a_n \sim 2\pi n / \log n$ relation to express $N_J(E)$ as a function of E .
- (3) *Trace error* (Theorem B.1): $O(1/\sqrt{N})$. This is the average error over all $k \leq N-1$.

The apparent disparity—eigenvalue error decays but trace error saturates at $\text{RMS}_\infty \approx 0.61$ —is explained by the k -dependence. For small k (near the boundary), the eigenvalue error is larger; for large k , it decays as $(\log k)^{3/2}/k$. The RMS average over all $k \leq N$ is dominated by the small- k regime, yielding

a non-zero limit. This is consistent: the trace bound is an aggregate measure, while the eigenvalue bound is pointwise.

Appendix C. Derivation of the Correction Formula

From the Riemann–von Mangoldt formula: $\theta(\gamma_k) = \pi(k-1-S(\gamma_k))$. From the Gram point definition: $\theta(g_{k-1}) = \pi(k-1)$. Subtracting: $\theta(\gamma_k) - \theta(g_{k-1}) = -\pi S(\gamma_k)$.

The Mean Value Theorem gives $\theta(\gamma_k) - \theta(g_{k-1}) = \theta'(\xi_k) \cdot (\gamma_k - g_{k-1})$ for some ξ_k between γ_k and g_{k-1} . Expanding $\theta'(\xi_k) = \theta'(g_{k-1}) + O(|\gamma_k - g_{k-1}|/g_{k-1})$ and solving for $\gamma_k - g_{k-1}$ yields the result. To derive the -0.5 shift: from the Riemann–von Mangoldt formula and the Gram point definition we have $\gamma_k - g_{k-1} = -\pi S(\gamma_k)/\theta'(g_{k-1}) + O(S^2/\theta')$.

The analytic diagonal satisfies $a_k - g_{k-1} = \pi/\log(g_{k-1}/2\pi)$. Since $\theta'(t) = \frac{1}{2}\log(t/2\pi) + O(1/t^2)$, we have

$$\log(g/2\pi) = 2\theta'(g) + O(1/g^2),$$

hence $\pi/\log(g/2\pi) = \pi/(2\theta'(g)) + O(1/(g \log g))$.

The residual is therefore

$$\gamma_k - a_k = -\frac{\pi S(\gamma_k)}{\theta'(g_{k-1})} - \frac{\pi}{2\theta'(g_{k-1})} = -\pi \frac{S(\gamma_k) + \frac{1}{2}}{\theta'(g_{k-1})} + O\left(\frac{1}{k \log k}\right).$$

The right-limit convention $S(\gamma_k^+)$ includes the jump of $\arg \zeta$ at the zero: $\arg \zeta$ jumps by $+\pi$ as T crosses γ_k from below, so $S(\gamma_k^+) = S(\gamma_k) + 1$. Substituting $S(\gamma_k) = S(\gamma_k^+) - 1$ into the residual gives

$$\gamma_k - a_k = -\pi \frac{S(\gamma_k^+) - \frac{1}{2}}{\theta'(g_{k-1})} + o(1),$$

which is the correction formula. The constant 0.5 arises from the conversion factor $\pi/\log = \pi/(2\theta')$ (contributing $-1/2$) and the right-limit jump convention $S(\gamma_k^+) = S(\gamma_k) + 1$ (contributing $+1$), whose net effect is the shift $-S(\gamma_k^+) + 1/2$. Empirical verification gives RMS 0.0090 over $N = 1000$ zeros (see Section 8).

Appendix D. Rigorous Weyl Law for J_N

Coefficient Asymptotics.

LEMMA D.1 (Gram Point Asymptotics). $g_n = \frac{2\pi n}{\log n} + O\left(\frac{n}{\log^2 n}\right)$.

Proof. From the Stirling expansion $g_n = 2\pi n / \log n + O(n / \log^2 n)$ (see §6, Definition 6.1). Newton iteration with $\theta'(t) = \frac{1}{2} \log \frac{t}{2\pi} + O(1/t^2)$ refines this to the stated accuracy. $\square$

LEMMA D.2 (Diagonal Asymptotics). $a_n = \frac{2\pi n}{\log n} + O\left(\frac{n}{\log^2 n}\right)$ for $n \geq 1$.

Proof. $a_n = g_{n-1} + \pi / \log(g_{n-1}/2\pi)$. By Lemma D.1, $g_{n-1} \sim 2\pi n / \log n$. The correction term $\pi / \log(g_{n-1}/2\pi) = O(1)$, absorbed into the $O(n / \log^2 n)$ error. $\square$

LEMMA D.3 (Off-Diagonal Asymptotics). $b_n = \sqrt{\frac{\pi}{2}} \sqrt{\log n} + O\left(\frac{1}{\sqrt{\log n}}\right)$. In particular $b_n = O(\sqrt{\log n})$.

Proof. $g_{n+1} - g_n = 2\pi / \log n + O(1 / \log^2 n)$ and $\theta'(g_{n+1}) = \frac{1}{2} \log n + O(\log \log n)$. Multiplying: $b_n = \sqrt{2\pi / \log n} \cdot \frac{1}{2} \log n = \sqrt{\frac{\pi}{2}} \sqrt{\log n} + O(1 / \sqrt{\log n})$. $\square$

COROLLARY D.4 (Ratio Decay). $b_n / a_n = O\left(\frac{(\log n)^{3/2}}{n}\right) \rightarrow 0$ as $n \rightarrow \infty$.

Gershgorin Localisation.

LEMMA D.5 (Gershgorin Bound). $|\lambda_k - a_k| \leq b_{k-1} + b_k = O(\sqrt{\log k})$, where $b_{-1} = b_{N-1} = 0$.

Proof. Standard Gershgorin theorem for tridiagonal matrices: disk k has centre a_k and radius $|b_{k-1}| + |b_k|$. Since the off-diagonals are strictly positive, the eigenvalues are simple and each lies in a distinct Gershgorin disk. The bound follows from Lemma D.3. $\square$

COROLLARY D.6 (Relative Error). $|\lambda_k - a_k| / a_k = O\left(\frac{(\log k)^{3/2}}{k}\right) \rightarrow 0$.

Proof. Divide the Gershgorin bound by $a_k \sim 2\pi k / \log k$. $\square$

Counting Function.

PROPOSITION D.7 (Eigenvalue Count). $N_J(E) = \#\{k : \lambda_k \leq E\} = n_E + O((\log E)^{3/2})$, where n_E is defined by $a_{n_E} \leq E < a_{n_E+1}$.

Proof. From Corollary D.6, $\lambda_k \leq E$ iff $a_k \leq E + O(\sqrt{\log k})$. The number of eigenvalues for which $|a_k - E| \leq b_{k-1} + b_k$ (the “uncertain” band) is at most $O(\sqrt{\log k}) / (2\pi / \log k) = O((\log k)^{3/2})$, using the level spacing $a_{k+1} - a_k \sim 2\pi / \log k$. $\square$

PROPOSITION D.8 (Inversion). $n_E = \frac{E}{2\pi} \log \frac{E}{2\pi} - \frac{E}{2\pi} + O\left(E \frac{\log \log E}{\log E}\right)$.

Proof. Set $E = a_n = 2\pi n / \log n + O(n / \log^2 n)$. Inverting $f(x) = 2\pi x / \log x$: $f^{-1}(y) = y \log y / (2\pi) + O(y \log \log y / \log y)$. Substituting $y = E$ gives the claim. $\square$

THEOREM D.9 (Weyl Law). $N_J(E) = \frac{E}{2\pi} \log \frac{E}{2\pi} - \frac{E}{2\pi} + O\left(E \frac{\log \log E}{\log E}\right)$.

Proof. Combine Propositions D.7 and D.8. For Theorem 6.4:

$$N_J(E) \sim \frac{E}{2\pi} \log E, \quad \text{error } o(E \log E).$$

Weak convergence of μ_N follows. $\square$

COROLLARY D.10 (Empirical Spectral Measure). $\mu_N = \frac{1}{N} \sum_{k=0}^{N-1} \delta_{\lambda_k^{(N)}}$ converges weakly to $\rho(E)dE$ with $\rho(E) = \frac{1}{2\pi} \log \frac{E}{2\pi}$.

Killip–Simon and Geronimo–Case Conditions.

LEMMA D.11 (Killip–Simon Sum Rule). $\sum_{n=1}^{\infty} (b_n/a_n)^2 < \infty$.

Proof. By Lemma D.3 and Lemma D.2, $(b_n/a_n)^2 = O((\log n)^3/n^2)$, which is summable. A direct computation over the first 200 terms gives $0.059 < 1$, well within the Killip–Simon threshold [9]. $\square$

LEMMA D.12 (Geronimo–Case Smoothness).

$$\sum_{n=2}^{\infty} |(b_n/a_n) - (b_{n-1}/a_{n-1})| < \infty.$$

Proof. The ratio follows $b_n/a_n \sim (\log n)^{3/2}/n$; the discrete derivative decays as $O((\log n)^{3/2}/n^2)$, whose sum converges. A direct sum over $n = 2$ to 200 gives 0.069. $\square$

These conditions, together with $b_n/a_n \rightarrow 0$, satisfy the hypotheses of Geronimo–Case [16] for purely absolutely continuous limiting spectrum on $[E_{\min}, \infty)$ with the stated Weyl density.

A fully detailed proof with all intermediate constants and integral bounds is available in the companion code `weyl_law_verify.c`: the Gershgorin bound is verified to 100% (50/50 eigenvalues satisfy), the spectral density ratio ρ_J/θ' is within 0.05% for $k \geq 10$, and the Killip–Simon and Geronimo–Case sums converge rapidly.

Appendix E. Sturm Oscillation and the Scattering Phase

LEMMA E.1 (Sturm Oscillation). *For the real symmetric Gram Jacobi matrix J_N ,*

$$\arg \det(J_N - EI + i0) = \pi N_J(E) \pmod{2\pi},$$

where $N_J(E) = \#\{k : \lambda_k \leq E\}$.

Proof. The characteristic polynomial $p_N(z) = \det(zI - J_N)$ has real coefficients and factors as $p_N(z) = \prod_{k=0}^{N-1} (z - \lambda_k)$. For $z = E + i\varepsilon$ with $\varepsilon \rightarrow 0^+$, each factor $(E + i\varepsilon - \lambda_k)$ has argument ≈ 0 when $E > \lambda_k$ and $\approx \pi$ when $E < \lambda_k$. Hence $\arg p_N(E + i0^+) = \pi \cdot \#\{k : \lambda_k > E\} = \pi(N - N_J(E)) \pmod{2\pi}$. Since $\det(J_N - EI) = (-1)^N \det(EI - J_N)$, the factor $(-1)^N$ contributes $N\pi$ to the argument, giving $\arg \det(J_N - EI + i0) = \pi N_J(E) \pmod{2\pi}$. $\square$

For the free Jacobi: $\arg \det(J_N^{\text{free}} - EI + i0) = \pi N_{\text{free}}(E) \pmod{2\pi}$ by the same argument. The determinant ratio gives the spectral shift:

$$\arg \frac{\det(J_N - EI + i0)}{\det(J_N^{\text{free}} - EI + i0)} = \pi(N_J(E) - N_{\text{free}}(E)) \pmod{2\pi}.$$

From the Weyl law (Theorem 6.4), $N_{\text{free}}(E) = \theta(E)/\pi + C + o(1)$. The Sturm oscillation count for J_N gives $N_J(E) \rightarrow N(E) = \theta(E)/\pi + 1 + S(E)$. The argument converges to $-\pi S(E) \pmod{2\pi}$.

Appendix F. Proof of the Spectral Determinant Identity and Riemann Hypothesis

With the trace formula established (Theorem 10.3), taking $h_z(E) = \log(1 - E/z)$ for fixed $z \in \mathbb{C} \setminus \mathbb{R}$, the trace formula gives

$$\text{Tr } h_z(J_N) - \sum_{n=0}^{N-1} h_z(\gamma_n) \rightarrow C[h_z],$$

where $C[h_z]$ is the Weil prime sum. The convergence is locally uniform in z on compact subsets of $\mathbb{C} \setminus \mathbb{R}$ (see Step 4 of the proof of Theorem 10.5 for the detailed justification).

The Hadamard product for ξ gives $\log \xi(\frac{1}{2} + iz)/\xi(\frac{1}{2}) = \sum_n \log(1 - z/\gamma_n)$ (converging under symmetric pairing $\rho \leftrightarrow 1 - \rho$). Since locally uniform convergence of logarithms implies locally uniform convergence of exponentials (the exponential function is locally Lipschitz), we exponentiate to obtain $\det(zI - J_N)/\det(z_0I - J_N) \rightarrow c \cdot \xi(\frac{1}{2} + iz)/\xi(\frac{1}{2} + iz_0)$.

Appendix G. Distributed Krein Self-Consistency

The spectral measure convergence underlying the determinant identity is established in the proof of Theorem 10.1 (Steps 2 and 5 of that proof). The following numerical data confirms the self-consistency of the distributed Krein shift.

The Hellman–Feynman theorem gives $\partial\lambda_k/\partial a_n = |v_n^{(k)}|^2$. The matrix $M_{kn} = \partial\lambda_k/\partial a_n$ (finite differences, $\Delta = 10^{-6}$, $N^2 = 10\,000$ solves) satisfies

$$M \cdot \delta a^{\text{corr}} = \delta\lambda + \varepsilon, \quad \text{RMS}(\varepsilon) = 0.809,$$

where $\delta\lambda_k = \lambda_k - \gamma_k$ and $\delta a_n^{\text{corr}} = -\pi(S(\gamma_n^+) - 0.5)/\theta'(g_{n-1})$. Row sums average 0.999998. The forward RMS 0.809 against eigenvalue errors $\delta\lambda$ (RMS 0.854) confirms the correction formula is the self-consistent solution.

Appendix H. The Gauss Sum Analogy

The distributed Krein shift has a natural interpretation as a Gauss sum:

Legendre symbol $\chi_p(a) = (a p) = \pm 1$	$S(T) = \frac{1}{\pi} \arg \zeta(\frac{1}{2} + iT)$
Gauss sum $G(\chi_p) = \sum \chi_p(a) e^{2\pi i a/p}$	Weil sum $\sum_{p,m} \frac{\log p}{p^{m/2}} \hat{h}(m \log p)$
DFT at $k = N/p$ reveals $G(\chi_q)$	Resolvent trace reveals $\frac{\xi'}{\xi}(\frac{1}{2} + iE)$
Finite sum over $\mathbb{Z}/p\mathbb{Z}$	Infinite sum over all primes

In both cases, a sum of “characters” weighted by oscillatory kernels produces spectral peaks at the characteristic frequencies of the underlying arithmetic object—revealing the prime distribution through the Fourier transform.

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